

Coping with growth and emigration: Latvian labor market before and after EU accession

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Abstract

In 2002-2005, Latvian labor market has witnessed dramatic changes related both to unprecedented economic growth and to a massive outflow of labor force after EU enlargement in May 2004. This study, based on micro-level analysis of Latvian Labor Force surveys 2002-2005, provides a detailed analysis of these changes, with a particular focus on developments which are likely to improve living standards of formerly disadvantaged segments of population. The findings suggest that emerging shortage of labor has led to strong wage growth and reduction of unemployment. The author documents increased employment rates and improvements in labor market position of ethnic minorities, elderly, fixed-term workers, the low-skilled, and other disadvantaged groups. Wage growth, in turn, has fostered labor force participation, resulting in further increase of employment rates. This analysis may be of interest for policy makers in countries entering a period of transformation or exposed to external shocks.

The study also provides an in-depth analysis of the labor market integration of the sizable Russian-speaking minority in Latvia. The author finds, on average, modest occupational segregation but somewhat larger segregation as far as the sectors of the economy are concerned. In both cases, segregation index is inversely related to Latvian language skills; the same is true for the observed ethnic wage gap. The unexplained ethnic wage gap reaches its maximum (6 percent) for workers with medium Latvian language skills.

JEL Classification: J21, J31, J64, J15

Keywords: employment; unemployment; ethnic minorities; returns to education; wage decomposition

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CONTENTS

INTRODUCTION	4
1. ACCESS TO JOBS	4
A. EMPLOYMENT DYNAMICS	4
B. THE ETHNIC GAP IN EMPLOYMENT IS NARROWING DOWN.....	8
C. CHANGES IN EMPLOYMENT AND PARTICIPATION PATTERNS.....	9
D. SCHOOL-TO-WORK TRANSITION AND YOUTH UNEMPLOYMENT.....	10
E. CHANGES IN UNEMPLOYMENT RISK FACTORS.....	11
F. EMPLOYMENT COMPOSITION.....	15
G. OCCUPATIONAL AND SECTORAL SEGREGATION.....	16
2. THE EARNINGS STRUCTURE	18
A. EARNINGS AND EDUCATION.....	18
B. EXPERIENCE AND EARNINGS.....	21
C. WAGE DISCRIMINATION AND SEGREGATION.....	25
D. REGIONAL EFFECTS.....	26
E. INDUSTRY WAGE DIFFERENTIALS.....	28
F. CHANGES IN OVERALL EARNINGS STRUCTURE.....	28
CONCLUDING REMARKS.....	30
REFERENCES	31

TABLES

Table 1.1: Employment rates (%) by ethnicity and demographic group, 2002-2005. Population aged 15-74.....	34
Table 1.2: Employment rates (%) by ethnicity and residence, 2002-2005. Population aged 15-74.....	34
Table 1.3: Determinants of employment of population aged 15-74, 2002 and 2005.....	35
Table 1.4: Determinants of labor force participation of population aged 15-74, 2002 and 2005.....	36
Table 1.5: Labor force participation rates (%) by ethnicity and demographic group. Population aged 15-74, 2002-2005.....	37
Table 1.6: Population aged 15-24 by participation in education and in the labor force, 2002 and 2005.....	38
Table 1.7: Determinants of unemployment. Population aged 15-74, 2002 and 2005.....	39
Table 1.8: Unemployment rates (%) by ethnicity and demographic group. Population aged 15-74, 2002-2005.....	40
Table 1.9: Employment status, incidence of part-time and temporary work, and wage distribution in the main job by ethnicity, 2002-2005.....	41
Table 1.10: Occupation in the main job by ethnicity, 2002 and 2005. Employees aged 15-74.....	42
Table 1.11: Occupation in the main job by Latvian language skills, 2005. Employees aged 18-64.....	43
Table 1.12: Industry in the main job by ethnicity, 2002 and 2005. Employees aged 15-74.....	44
Table 1.13: Sector of economic activity in the main job by Latvian language skills, 2005. Employees aged 18-64.....	45
Table 2.1: Full-time employees by educational attainment, 1997-2005.....	46
Table 2.2: Estimated <i>ceteris paribus</i> wage differentials (%) associated with educational attainment. Full-time employees, 2005 and 2002.....	46
Table 2.3: Estimated <i>ceteris paribus</i> wage differentials associated with job tenure. Full-time employees, 2005.....	47
Table 2.4: Incidence of fixed-term contracts and estimated <i>ceteris paribus</i> wage differentials between temporary and permanent workers. Full-time employees, 2005 and 2002.....	47
Table 2.5: Gross male – female wage differentials and productivity differentials. Full-time employees, 2005 and 2002.....	48

Table 2.6: Gross wage differentials and productivity differentials between ethnic Latvian and non-Latvian workers. Full-time employees, 2005 and 2002.....	48
Table 2.7: Gross wage differentials and productivity differentials between native Latvian speakers and other workers, by self-reported Latvian language skills level. Full-time employees aged 18-64, 2005.....	49
Table 2.8: Estimated <i>ceteris paribus</i> regional effects on earnings. Full-time employees, 2005 and 2002.....	49
Table 2.9: Estimated <i>ceteris paribus</i> regional wage differentials (%). Full-time employees, 2005.....	50
Table 2.10: Estimated <i>ceteris paribus</i> regional wage differentials (%) when local unemployment rate is controlled. Full-time employees, 2005.....	50
Table 2.11: Estimated <i>ceteris paribus</i> industry effects on earnings. Full-time employees. 2005 and 2002.....	51
Table 2.12: Changes in earnings structure of full-time workers between 2002 and 2005: decomposition analysis.....	52

FIGURES

Figure 1.1: Employment rates in selected EU countries: 2000, 2002, and 2005.....	5
Figure 1.2: Change in employment rates in selected EU countries: 2005 vs. 2000 and 2002.....	6
Figure 1.3: Employment, labor force participation, and unemployment rates by gender and age group: 2005 vs. 2002.....	7
Figure 1.4: Employment rates by educational attainment and ethnicity.....	8
Figure 1.5: Youth unemployment in Latvia, EU 15, and new member states, 1998-2005.....	11
Figure 1.6: Reduction in predicted risk of unemployment between 2002 and 2005 by region and educational attainment.....	12
Figure 1.7: Predicted age-unemployment risk profiles, 2002 and 2005	14
Figure 2.1: Returns to education in Latvian labor market. Full-time employees, 2005.....	19
Figure 2.2: Returns to education in Latvian labor market by source and group of employees, 2005.....	20
Figure 2.3: Estimated age – earnings profiles of full-time employees, 2005.....	22

BOXES

Box 1: “Access” vs. “Reward” Wage Differentials.....	20
Box 2: Detailed wage decomposition.....	29

INTRODUCTION

This study was undertaken in the framework of Latvia Living Standards Assessment 2006. There are many ways in which labor market affects living standards of different groups of population: employment and unemployment rates; job quality and stability; overall wage level; returns to education and experience; regional wage differentials; gender and ethnic wage gaps. In 2002-2005, Latvian labor market has witnessed dramatic changes related both to unprecedented economic growth (28 percent in 3 years) and to massive outflow of labor force after EU enlargement in May 2004. Emerging shortage of labor has led to strong growth of real wages and reduction of unemployment. Moreover, it has increased employment rates and improved labor market position of ethnic minorities, females with children, elderly, fixed-term workers, and other disadvantaged groups. Wage growth, in turn, has increased labor force participation, resulting in further increase of employment rates. Importantly, these changes were much stronger and faster than expected by policy makers.

In what follows we provide a detailed analysis of these changes, with a particular focus on developments which are likely to improve living standards of formerly disadvantaged segments of population. This analysis may be of interest for policy makers in many countries which are entering a period of transformation.

1. ACCESS TO JOBS

A. EMPLOYMENT DYNAMICS

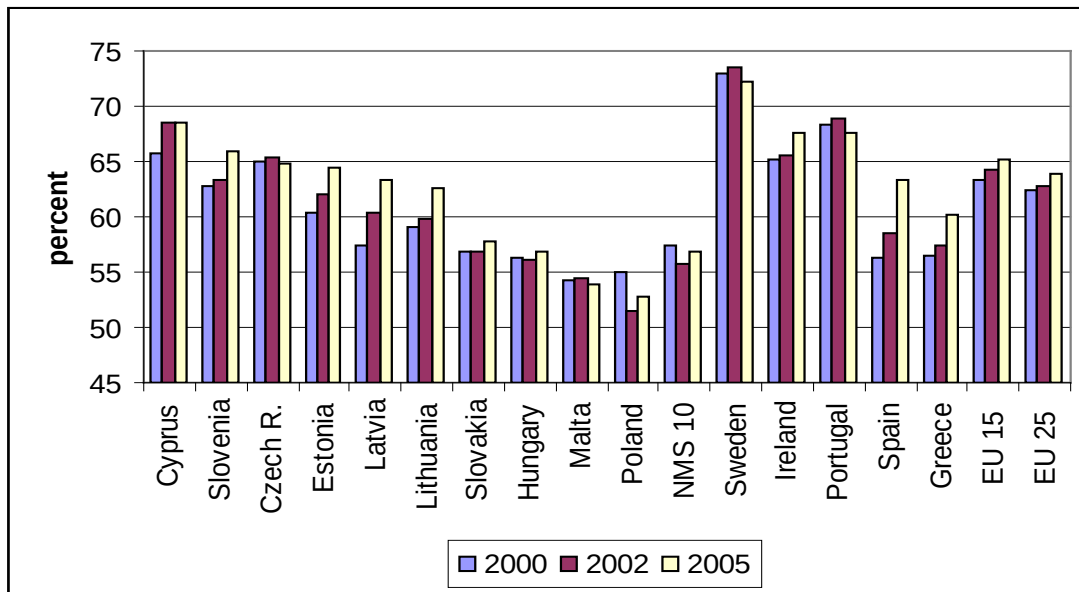
Growth in employment rates in Latvia, which started in 2001, went on in 2002-2005. Here we focus on the latter period, immediately before and after the EU enlargement (trends in Latvian labor market in 1997-2002 are described in Hazans 2005). Several factors have contributed to changes in employment rates: outward shift in labor demand caused by economic growth; increase in labor force participation due to increase in minimal wages in 2003 and 2004 and deductibles in 2005, as well as overall strong wage growth in 2004-2005; increase of labor force participation by the elderly due to raise in legal retirement age of men (2003) and women (2003, 2004, and 2005).

The subsequent discussion is entirely based on results of Latvian Labor Force Surveys 2002-2005. Hence all estimates are subject to measurement errors related to temporary or permanent emigration of significant part of Latvian labor force in 2004-2005. As long as temporary migrants (which are mostly employed but heavily under-represented in LFS) remain part of Latvian population, LFS estimates of employment and participation rates are biased downward. On the other hand, absolute majority of migrants (both temporary and permanent) are not excluded from the population register, which is used as sampling frame for the LFS; this to some extent distorts sampling weights, as long as permanent migrants (a group which is of course different from total population in terms of age and other characteristics) should have been in fact excluded from the register. This problem is

common for all countries exposed to substantial emigration but is rarely acknowledged. Estimating measurement errors is not attempted here.

By 2005, as seen in Figure 1.1, Latvia's employment rate of population aged 15-64 was 63.4 percent, quite close to the EU-25 average of 63.8 percent and even EU-15 average of 65.1 percent, although somewhat lower than in Estonia (64.4), Czech Republic (64.8), Slovenia (66.0), and Cyprus (68.5). However, both in 2000-2005 and in 2002-2005, Latvian employment rate has been rising faster than that of any other new EU member state and also faster than that of any other EU member except Spain (Figure 1.2).

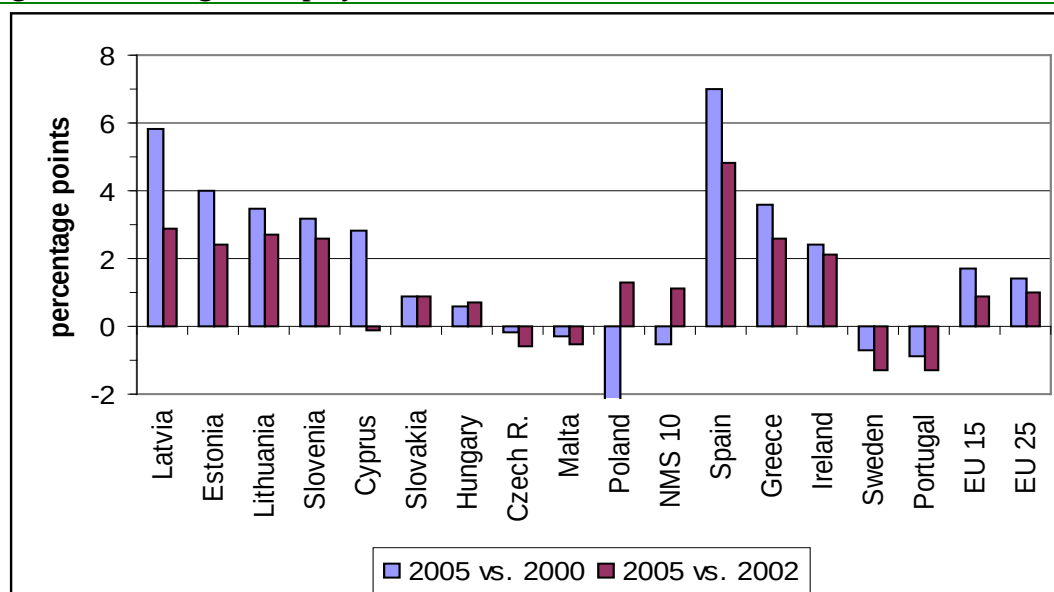
Figure 1.1: Employment rates in selected EU countries: 2000, 2002, and 2005.



Notes: Employment rates for population aged 15 to 64. The chart includes all new member states, as well as those old member states where employment rates experienced the largest increase (Spain, Greece, and Ireland) or decline (Sweden, Portugal).

Source: Eurostat.

Figure 1.2: Change in employment rates in selected EU countries: 2005 vs. 2000 and 2002



Notes: Employment rates for population aged 15 to 64. The chart includes all new member states, as well as old member states where employment rates experienced the largest increase (Spain, Greece, Ireland) or decline (Sweden, Portugal). *Source:* Calculation based on Eurostat data.

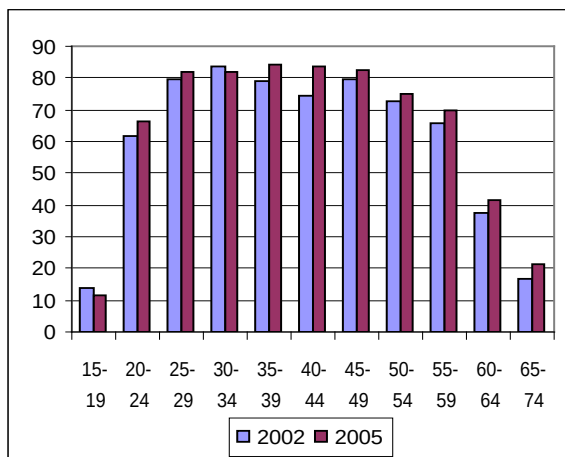
Overall unemployment rate of population aged 15 to 74, which as high as 12 percent in 2002, went down to about 10.5 percent in 2003-2004; a further decline to 8.7 percent in 2005 was at least in part due to outflow of the labor force abroad after EU enlargement.

Figure 1.3 displays evolution of the key labor market indicators by gender and age group between 2002 and 2005. Increase in employment rates has been especially strong for 35-44 and 55-64 years old men and women, as well as for men aged 20-24 and 65-74 (Figure 1.3). The factors which contributed to increase in employment rates differ across demographic groups. Higher employment of men aged 20-24 has resulted exclusively from reduction in unemployment rate. Men aged 35-44 saw a very substantial reduction in unemployment rate combined with a modest increase in participation. Men aged 45-54 saw a fall in unemployment rate from 11.7 to 9.9 percent and a slight increase in participation. Older men have benefited mainly from increase in participation (by 4 to 5 percentage points), but also from a reduction in unemployment (by one percentage point for the 55-64 year olds and by two points for 65-74 year olds).

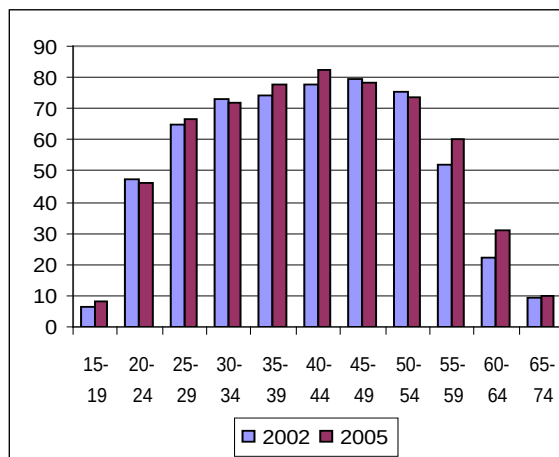
Women aged 35-44 experienced reduction in unemployment rate from 10.4 to 6.4 percent without significant change in participation. Increase in employment rates of women aged 55-64 was ensured by a huge increase in participation (from 38.7 to 47.9 percent) combined with a reduction of unemployment rate from 7.9 to 6.7 percent.

Figure 1.3: Employment, labor force participation, and unemployment rates by gender and age group: 2005 vs. 2002

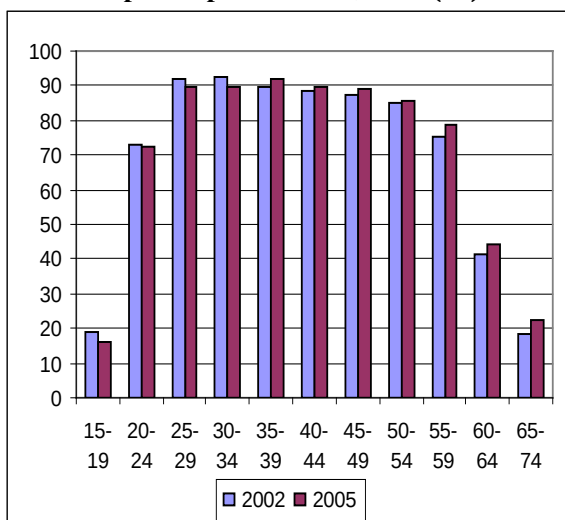
Employment rates, men (%)



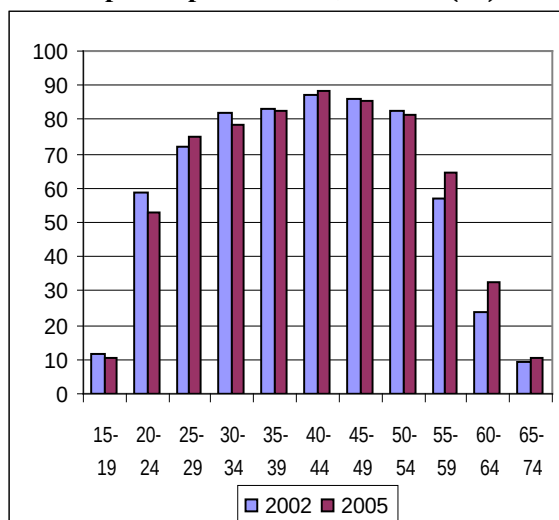
Employment rates, women (%)



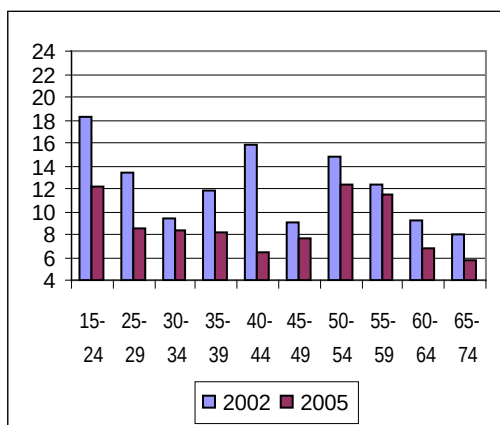
LF participation rates, men (%)



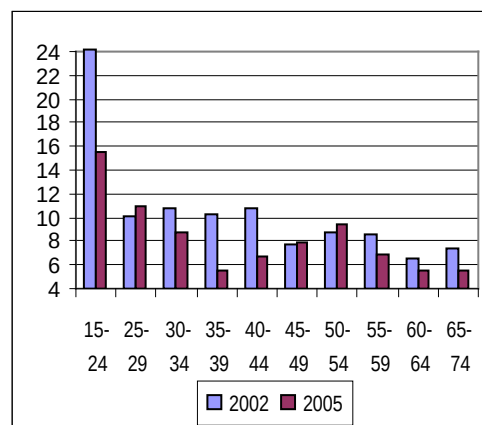
LF participation rates, women (%)



Unemployment rates, men (%)



Unemployment rates, women (%)



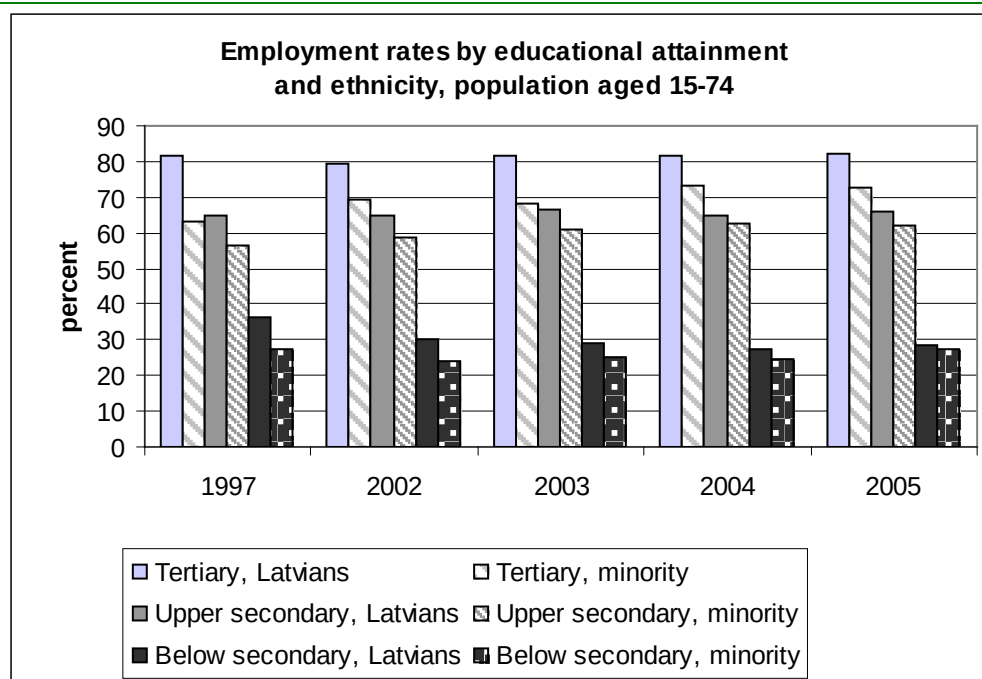
Source: Calculation based on LFS data.

B. THE ETHNIC GAP IN EMPLOYMENT IS NARROWING DOWN

While minority population still has somewhat lower employment rate, by 2005 the overall gap in the age group 15-74 has been reduced from more than 6 percentage points in 2002 to less than 3 percentage points in 2005 (Table 1. 1). The increase in employment between 2002 and 2005 has taken place primarily (and as far as women are concerned, exclusively) within minority population. The ethnic gap in employment rates has narrowed down in all age groups except the youth and the elderly. For the youth, however, the difference in rates is not statistically significant in any of the years under inspection.

The ethnic gap in employment remains substantial (10 percentage points) for persons with tertiary education. While this is just half of the gap width in 1997 (see Figure 1.4), there have been no significant change in this respect since 2002.

Figure 1.4: Employment rates by educational attainment and ethnicity



Source: Calculation based on LFS data.

Inspection of employment rates by type of settlement (Table 1.2) reveals a substantial reduction in the ethnic employment gap in both rural and urban areas, especially outside Riga. However, in all four types of populated areas employment rates of non-Latvians lags behind those of Latvians by four to five percentage points. The overall gap is smaller only due to higher concentration of non-Latvians in Riga, where employment rate is higher than elsewhere.

C. CHANGES IN EMPLOYMENT AND PARTICIPATION PATTERNS

As was shown in the previous section, although overall [observed] labor force participation rates has hardly changed between 2002 and 2005, there were substantial changes for some demographic groups. As far as employment is concerned, observed rates have changed as well. Table 1.3 (respectively, 1.4) compares determinants of employment (respectively, labor force participation) in 2002 and 2005.

The most noticeable change is perhaps the fact that for men, the *ceteris paribus* ethnic gaps in participation and employment have both disappeared. For women, the *ceteris paribus* ethnic participation (respectively, employment) gap has shrunk from 7 to 3.5 (respectively, from 9 to 6) percentage points. See Table 1.5 for details of the evolution of participation rates by ethnicity and demographic group.

Other things equal, women aged 20-24 or 45-54, as well as women with children below 15 years of age, have become less active and less likely to be employed. By contrast, women aged 55-64 have become relatively more active in the labor market and slightly more employable. All these findings are consistent with changes in observed rates.

Men aged 35-44 are still less likely to participate in the labor force and to be employed than their otherwise similar counterparts aged 25-34, but the difference between these two cohorts has become smaller, especially in terms of employment. Moreover, in contrast with 2002, the observed employment and participation rates in 2005 are even slightly higher for the older cohort (see Figure 1.3).

In 2002, men and women in Kurzeme were significantly less active and less likely to be employed than their otherwise similar counterparts in Vidzeme and Riga region; this difference disappeared by 2005.

Table 1.5 looks more closely at the evolution of the ethnic gap in participation. Catching up took place both for low-skilled workers (where participation has decreased for Latvians but increased for non-Latvians), and for persons with tertiary education (here activity of both ethnic groups has increased, but more so for the minority population). For men the observed participation gap (like the estimated *ceteris paribus* gap) has disappeared, but actually minority women have done even more catching up: the gap has decreased by 3 percentage points for women, while by only half of this amount for men.

The strongest reduction of the ethnic gap in participation has occurred in the age groups 25-34 (from eight percentage points to zero), 45-54 (from more than six to less than two percentage points), and 55-64 (from eleven to less than two percentage points). On the other hand, the gap has stayed unchanged for the elderly, while for the youth it went up from minus two to almost three percentage points. Plausibly, this latter phenomenon also manifests improving living conditions of the minority population: more families are able to support studies of their children.

D. SCHOOL-TO-WORK TRANSITION AND YOUTH UNEMPLOYMENT

Table 1.6 provides a simple descriptive view on school-to-work transition of the youth. Between 2002 and 2005, proportion of economically inactive students among population aged 15 to 24 has increased from 53.2 to 55.8 percent at the expense of all other categories (non-students and economically active students). Proportions of both unemployed and discouraged (ready to work but not looking for a job) persons among the youth have declined. These patterns are observed for both genders (these results are not shown), as well as for both ethnic groups, but they are more pronounced among non-Latvians. Such trends are consistent with rising living standards.

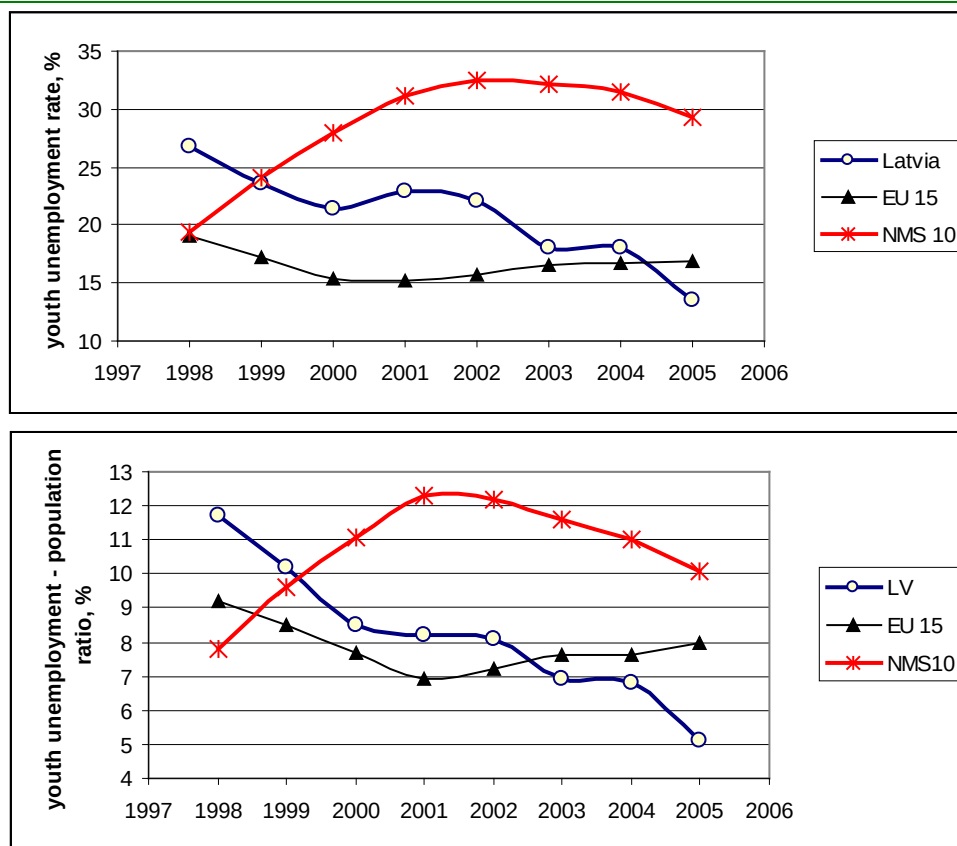
Overall proportion of students among the youth has increased from 63.0 to 64.8 percent. This increase has occurred almost exclusively within minority population, where total enrollment has gone up by 4.5 percentage points (estimated secondary and tertiary enrollment rates of non-Latvians aged 15 to 24 have increased by 2.6 and 4.0 percentage points respectively, while primary enrollment rate went down by 3.2 points).

Labor force participation rate of the minority youth has fallen from 40.6 to 35.5 percent, leading to a decrease in overall youth activity rate from 39.0 to 37.4 percent (despite a slight increase in labor force participation by young ethnic Latvians).

In Latvia, as elsewhere, unemployment rate among the youth is higher than in other age groups (Figure 1.3). However, both trend and level of youth unemployment, especially as compared to other European countries, suggest that this is not a major problem. In 2005, the average unemployment rate in the age group 15-24 was 30 percent in the 10 new EU member states (NMS-10), 17 percent in EU-15, and about 13 percent in Latvia (in Estonia and Lithuania this rate was 16 percent). During the last three years this indicator has declined on average by three percentage points in NMS-10, but increased by the same amount in EU-15, while in Latvia it has dropped by ten percentage points. Compared with 1998, youth unemployment rate has halved in Latvia, while its average level in NMS-10 (driven by the trends in Poland, Czech R., Slovakia, and Hungary) has increased substantially (Figure 1.5, upper panel).

Given the low level of economic activity among the youth, it is common to use for this age group unemployment-population ratio rather than unemployment rate (see e.g. Jouhette and Romans 2006). This indicator suggests even more convincingly that youth unemployment in Latvia is not a serious problem (see Figure 1.5, lower panel): just five percent of Latvian youth were unemployed in 2005; this has to be compared to eight percent in EU-15 and to ten percent in NMS-10. According to this indicator, Latvia performs better than EU-15 already since 2003. In 2005, only three EU members (Lithuania, Luxemburg, and Ireland) have done better than Latvia on this account. Even total proportion of unemployed and discouraged persons among the Latvian youth is lower than proportion of unemployed among the youth in NMS-10.

Figure 1.5: Youth unemployment in Latvia, EU 15, and new member states, 1998-2005



Source: Eurostat.

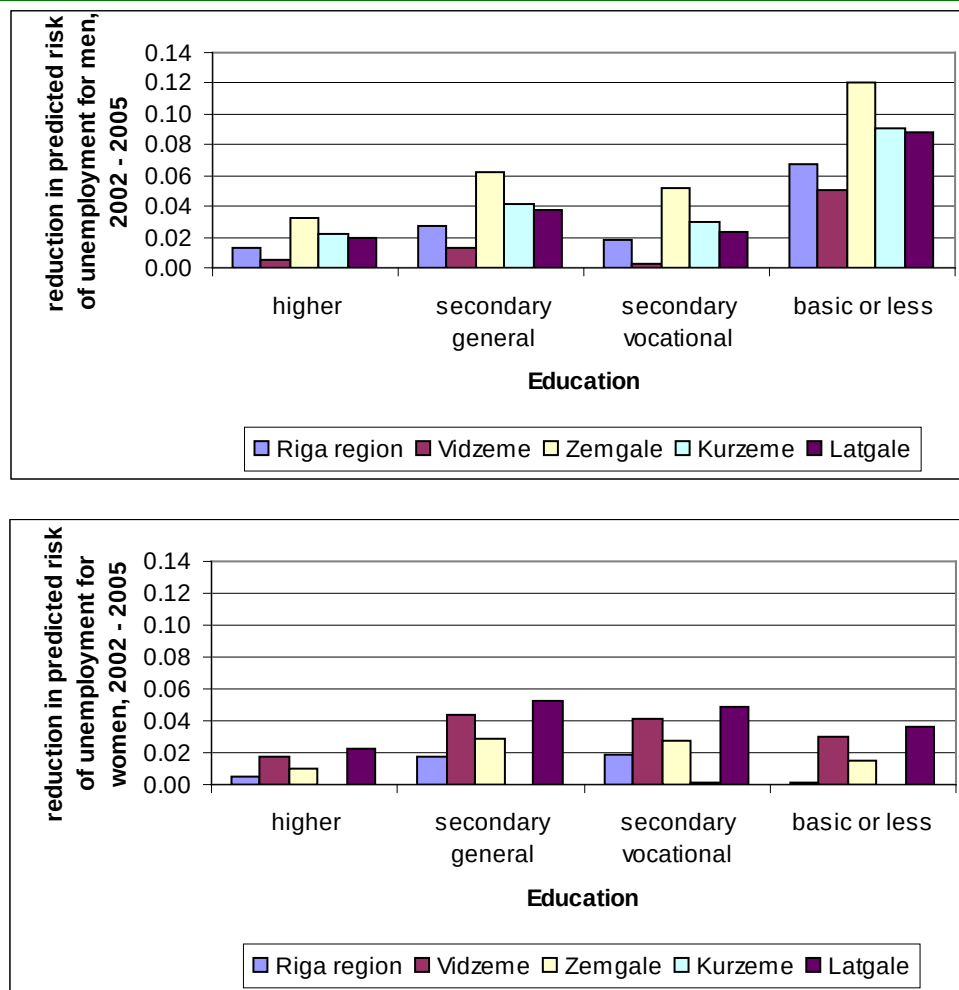
E. CHANGES IN UNEMPLOYMENT RISK FACTORS

As shown in Figure 1.3 (lower panel), since 2002 unemployment rates have declined significantly in all age and gender groups except females aged 45-54. Table 1.7 compares the determinants of unemployment in 2002 and 2005. Given that the focus of this study is the labor market outcomes of different socio-economic groups, we do not control explicitly for demand side factors (their impact is reflected in the regional effects), as well as for previous experience. The results do not change qualitatively when NUTS 3 (33 districts and main cities) level dummies are included instead of regional dummies.

Evolution of regional and skills effects is summarized in Figure 1.6. For men, the largest reductions of unemployment risk have occurred in Zemgale, Kurzeme, and Latgale (in this order). For men with less than upper secondary education predicted unemployment risk dropped on average by twelve percentage points in Zemgale, by nine points in Kurzeme and Latgale, by seven points in Riga region, and by five points in Vidzeme. Men with secondary general education have seen risk reduction by six points in Zemgale, by four points in Kurzeme and Latgale, and by three points in Riga region. Nevertheless even in 2005 the level of predicted unemployment risk remained high for

men with basic education (from 8 percent in Zemgale to 12-16 percent in Riga region, Kurzeme, and Vidzeme to 21 percent in Latgale), as well as for men with secondary general education in Kurzeme (8 percent), Vidzeme (10 percent) and Latgale (14 percent).

Figure 1.6: Reduction in predicted risk of unemployment between 2002 and 2005 by region and educational attainment.
Upper panel: men; Lower panel: women



Notes: Results are derived from probit model presented in Table 1.7. Other controls include ethnicity, age group, family status, and type of residential area (urban or rural). Variables not shown in the Figure are fixed at their mean values for each year and gender. For women with basic education in Kurzeme, predicted unemployment risk has increased by 3 percentage points (this is not shown). *Source:* Calculation based on LFS data.

Among women, the risk reduction was also substantial (except in Kurzeme region), although much smaller than among men: the highest risk reduction was observed for those with secondary general education (five percentage points in Latgale, four points in Vidzeme, three points in Zemgale). For women with secondary vocational and basic

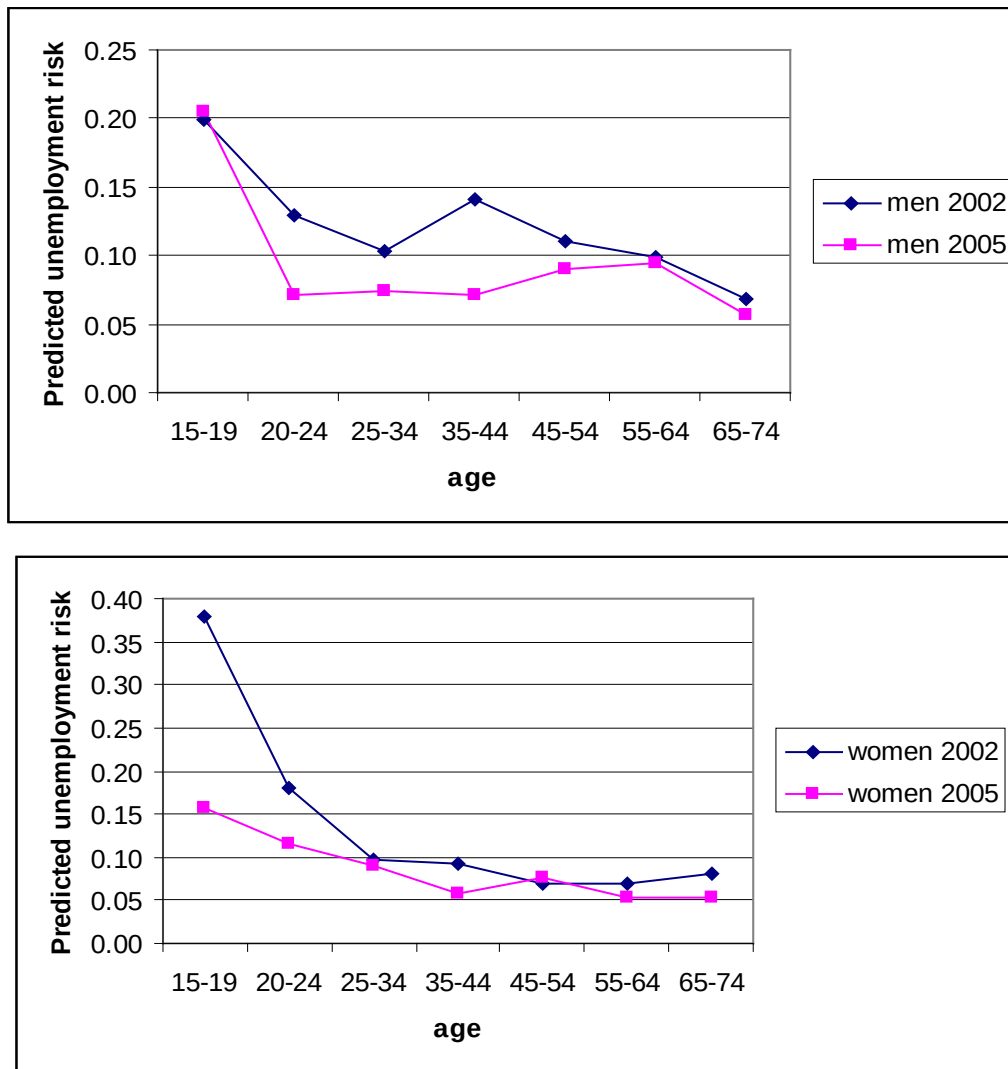
education, the reduction was in the range from four to five points in Latgale, three to four points in Vidzeme, and two to three points in Zemgale (Figure 1.6, lower panel). Like for men, predicted unemployment risk in 2005 was still high for low-skilled women (19 percent in Kurzeme and 12 to 15 percent in other regions), as well as for those with secondary general education (14 percent in Kurzeme and 9 to 11 percent elsewhere).

Ethnicity and unemployment risk. In 2002, the unemployment risk for non-Latvians was significantly higher than for otherwise similar Latvians: the unexplained ethnic gap was four percentage points for men and five percentage points for women. By 2005, this gap has disappeared for men but remained almost unchanged for women (Table 1.7). Table 1.8 provides a more detailed account of the evolution of the raw ethnic gap in unemployment. Unemployment rates have declined substantially for both ethnic groups and both genders, but more so for non-Latvians. In 2003, male unemployment rates dropped in both ethnic groups, while female unemployment declined only for Latvians. In 2004-2005, the difference between unemployment rates of non-Latvians and Latvians has declined by 2.7 points for men and by 2.6 points for women.

The ethnic gap in unemployment has narrowed in 2004 for low-skilled labor and in 2005 for medium-skilled labor. Among highly skilled individuals, however, unemployment has declined only for ethnic Latvians, and the gap has widened. By 2005, unemployment rates have declined in all age-ethnicity groups, except Latvians aged 55-64 and non-Latvians aged 45-54, while the ethnic gap has declined in all age groups except 45-54. For individuals aged 45 and more, however, the difference between unemployment rates among non-Latvians and Latvians remains substantial: about five percentage points.

Age-unemployment risk profiles. As follows from Table 1.7, when family status is controlled for, male age-unemployment risk profile (teenagers aside) was concave (inverse U-shaped) in 2002 but has become virtually flat in 2005. Female profile is declining with age (other things equal), but the profile has also become flatter in 2005 than in 2002. This is another confirmation of the fact that unemployment risk has become more evenly spread between social groups as a consequence of high growth and improved labor mobility. Family status, however, is strongly correlated with age. Hence a clearer picture of the full effect of age on unemployment risk can be obtained from a model which does not include dummies for being single and living with children below 15 years of age. Age-unemployment risk profiles derived from such model are presented in Figure 1.7. The main conclusion remains the same: In 2005, the male profile is flat in the age range from 20 to 44 years and increases only slightly in the range from 45 to 64 years. For females, unemployment risk declines with age in the range from 15 to 44 and is almost flat thereafter, with a slight peak between 45 and 54 years.

Figure 1.7: Predicted age-unemployment risk profiles, 2002 and 2005.
Upper panel: men; Lower panel: women



Notes: Results are derived from probit models. Other controls include educational attainment, ethnicity, region, and type of residential area (urban or rural). Variables not shown in the Figure are fixed at their mean values for each year and gender. *Source:* Calculation based on LFS data.

F. EMPLOYMENT COMPOSITION

While employment as such is an important determinant of living standards, type of employment plays its role, too. Table 1.9 compares distribution of employed Latvians and non-Latvians in 2002-2005 by employment status and working time (full-time vs. part-time), as well as type of contract and monthly earnings (the two latter comparisons are presented only for hired employees).

In general, distributions by employment status for the two ethnic groups are very similar (and become more similar over time): 84 (in 2002) to 87 (in 2005) percent of employed Latvians and about 90 percent of their minority counterparts are employees (i.e., salaried workers), 3 to 4 percent in each group are employers. About 7 percent of Latvians and 4 percent of non-Latvians are self-employed. Between 2002 and 2005, proportion of unpaid family workers (this category is found predominantly in farms) has decreased from 5 to 2.5 percent among Latvians and from almost 3 to 1.6 percent among employed minority population.

Incidence of part-time work, low by international standards, has further decreased between 2002 and 2005 from 13.5 to 9.0 percent among Latvians and from 11.8 to 7.2 percent among non-Latvians. In the same period, incidence of involuntary part-time employment has decreased by one third and in 2005 was about 3 percent in each ethnic group. Overall, proportion of part-time workers among employed population (8.3 percent) is just half of that in EU-15.

Incidence of fixed-term contracts, especially those of duration more than 6 months, has dropped dramatically during the last three years. In 2002, 14 percent of all employees had temporary contracts at their main jobs; 60 percent of these contracts were of duration longer than 6 months. In 2005, main job was temporary only for 8.4 percent of employees (8.6 percent among Latvians and 8.2 percent among minority workers), and only half of their contracts (53% for Latvians and 44% for non-Latvians) were for more than 6 months. While only among new EU members only Poland, Slovenia and Cyprus have higher incidence of temporary jobs than Latvia, on average both in EU-25 and in EU-15 it is much higher (close to 14 percent).

The above discussion suggests that incidence of part-time and temporary work is fairly similar among the two ethnic groups. Incidence of minimum wage is quite similar, too: (12.7 and 11.7 percent in 2005 among Latvian and non-Latvian workers, respectively). However, among minority employees a significantly higher proportion (one quarter vs. one fifth among Latvians) reports net monthly earnings between minimum and Ls 100, and significantly lower proportion (15.9 vs. 21.5 percent) reports earnings above Ls 200. A full analysis of the impact of different factors on earnings is postponed until chapter 2. Next several paragraphs examine the extent of segregation of the two ethnic groups among occupations and sectors of economic activities.

G. OCCUPATIONAL AND SECTORAL SEGREGATION

Table 1.10 compares occupational distribution of ethnic Latvian employees with distribution of their minority counterparts in 2002 and in 2005. It appears that these distributions are fairly stable over time. The proportion of highly skilled non-manual workers has slightly decreased in both ethnic groups at the expense of low-skilled non-manuals, more so among non-Latvians. Overall degree of occupational segregation is measured by *dissimilarity index DI* (known also as *Duncan index*); it is a number between 0 and 100, with 0 indicating equal distribution of ethnic (or other) groups among occupations, and 100 indicating complete segregation; see Ehrenberg and Smith 2005: p.396) In the given context, *DI* shows the minimal proportion (in percent) of non-Latvians which would have to change occupations in order to make their occupational distribution identical to that of Latvians'. This measure suggests a modest vertical segregation between ethnic groups: when the four "super-groups" (highly skilled non-manual, low-skilled non-manual, skilled manual, and elementary occupations) are considered, $DI_4 = 11.2$ (one percentage point up compared to 2002), while for the nine major groups of occupations $DI_9 = 12.4$. When men and women are considered separately, it appears that vertical segregation is more pronounced among women ($DI_4=14.7$) than among men ($DI_4=9.9$).

Latvians are in higher proportions in highly skilled non-manual occupations (especially the first two groups), while non-Latvians – in skilled manual and elementary occupations. In 2005, 22.6 percent Latvian employees were senior officials, managers or professionals, while among non-Latvians this proportion was 15.5 percent. It is worth noticing, however, that proportion of persons with tertiary education is also higher among ethnic Latvians than among minorities: 25.8 vs. 20.8 percent when all employees are considered and 25.1 vs. 16.1 percent for employees younger than 35 years. Hence promotion of tertiary education among ethnic minorities can be considered as a way of improving both their living conditions in a long-term perspective and the productive potential of Latvian labor force in general.

When both vertical and horizontal segregation is considered (by using 27 groups of occupations corresponding to two-digit codes), the dissimilarity index between Latvian and non-Latvian employees reaches 16.5, which also is not a high value.

Table 1.11 finds that occupational segregation is to a large extent explained by Latvian language skills. According to self-reported classification in a survey conducted between November 2005 and January 2006, 62.2 percent of employees aged 18 to 64 are native Latvian speakers, 20.1 percent have good knowledge of Latvian language, 12.2 percent – medium level knowledge, and 5.5 percent – poor knowledge. Among employees with good Latvian language skills just 7 percent would have to change occupation to make their occupational distribution identical to that of native Latvian speakers', while for workers with medium and poor Latvian language skills this proportion is, respectively, 25 and 49 percent.

Table 1.12 presents distribution of ethnic Latvian and minority employees among sectors of economic activity in 2002 and 2005 (for the latter year also with break-down by gender). Distributions of the two ethnic groups among sectors is much less similar than occupational distribution: index of dissimilarity calculated for the four “large” sectors (agriculture, industry, market services and non-market services) in 2005 was 19.5, while it was just 11.2 in the case of the four “large” groups of occupations; moreover, dissimilarity index for the four “large” economic sectors exceeds also dissimilarity index for the 27 two-digit groups of occupations.

Switching from the four “large” sectors to fourteen main sectors (according to NACE) results only in small increase of dissimilarity index. This suggests that the main difference between the majority and minority employees groups is indeed in the way how they are distributed among agriculture, industry, market and non-market services. More than one third of Latvians and just one out of five non-Latvians work in non-market services; almost nine percent of Latvians and just four percent of non-Latvians are employed in agriculture (recall that only hired employees are considered here). On the other hand, one out of three minority workers are found in industry, as opposed to one out of four Latvians. Market services employ two fifths of minority workers but less than one third of Latvians. It should be noticed also that in 2005, 38 percent of Latvian employees were employed in the public sector, while for minority employees this proportion was just 26 percent.

The picture changes dramatically when Latvian language skills are accounted for (see Table 1.13). Dissimilarity between minority employees with good knowledge of the state language and native speakers is modest: just 9 percent of either group would need to change sector of employment in order to have both groups distributed among agriculture, industry, market services, and non-market services in the same proportions. For workers with medium and poor Latvian language skills the dissimilarity index is, respectively, 15 and 22 percent. Switching to fourteen sectors increases each of the three dissimilarity indices by a factor of 1.5.

Distribution of employees among the fourteen main sectors is less stable than distribution among the nine main groups of occupations. Since 2002, the shares of construction and market services (especially real estate and business activities) have increased, while proportions of workers employed in agriculture, manufacturing and health have gone down. The dissimilarity index has increased by two and a half points in three years. Ethnic dissimilarity indices for men and women are roughly of the same magnitude.

2. THE EARNINGS STRUCTURE

One of the most important features of a labor market is its wage structure. Are human capital related workers' characteristics – education, general, and firm specific experience - rewarded in Latvia in similar ways as in established market economies? Empirical evidence from other post-communist countries suggests that returns to education have increased sharply, while returns to experience have decreased compared to the socialist wage setting system. However, the patterns of change and the end result differ across countries, reflecting differences in historical background, social norms, and the industrial structure of the economy.

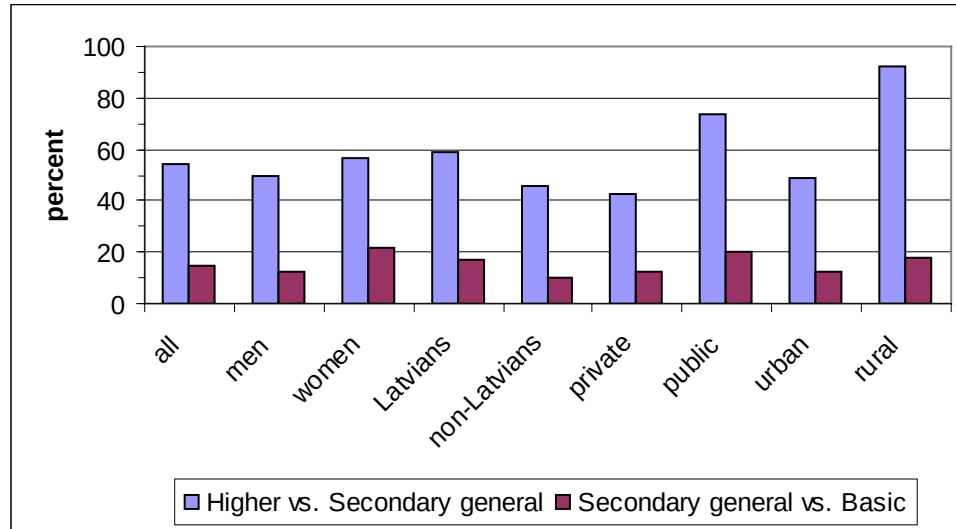
The following section looks at the major factors contributing to wage differences between individuals in Latvia and their evolution since 2002. Particular attention is focused on differences in wages earned by individuals of differing education levels, gender and ethnic wage differentials, and regional differences in wages. Such an assessment is of great importance in designing policies which target individuals and regions which may have been disadvantaged in the move to a market-based economy. Wage data in Latvian LFS are in the interval form, so the analysis is performed with the interval regression method (see Amemiya 1973; Long 1997).

A. EARNINGS AND EDUCATION

The issue of returns to education is of particular interest for Latvia, where the labor force is increasingly well-educated, as shown in Table 2.1. Results of this section are based on two earnings functions' specifications estimated from LFS 2002 and 2005. These functions account both for observable characteristics of individuals, and those of the firms in which they are employed. One specification, however, does not control for occupation and plant size, while the other does. Comparison of these two sets of results helps to understand how human capital is rewarded in the labor market, and also gives useful insights related to gender and ethnic issues.

Table 2.2 presents estimated *ceteris paribus* wage differential for full-time employees working at least 35 hours per week. By the year 2005, total returns to higher (tertiary) vs. general basic education in the Latvian labor market amounted to 76 percent on average. This implies that, on average, an individual with higher education earns 76 percent more of what an individual with basic education earns. These returns were significantly larger for females than for males (90 vs. 67 percent), for Latvians compared to non-Latvians (85 vs. 60 percent), in the public sector than in the private sector (110 vs. 59 percent), and for those working in the countryside compared to urban employees (126 vs. 66 percent). A similar pattern is found when returns to higher vs. secondary general education (54 percent on average) are considered. Figure 2.1 illustrates.

Figure 2.1 Returns to education in Latvian labor market. Full-time employees, 2005



Source: Calculation based on LFS data

Returns to secondary education (either general or vocational) are modest: both on average and for most groups they are in the range between 10 and 20 percent, but for women, as well for the public sector and rural employees – between 20 and 30 percent (in the rural case this only applies to vocational secondary education).

As noticed in Hazans (2005) based on year 2002 data, by international standards returns to tertiary education in Latvia are high, while returns to secondary education are low. This remains to be true. Returns to secondary education are particularly low for non-Latvians and in the private sector (recall that three quarter of minority employees are concentrated in the private sector, while for Latvians this proportion is 62 percent).

When different types of upper secondary vocational education are distinguished it appears that in 2005 wage differential between postsecondary vocational and general basic education was 22 percent for men and 31 percent for women, differential between secondary professional education and general basic education was 18 percent for men and 21 percent for women, while vocational education after basic education had no return for men and 11 percent for women (the latter type of education is not widespread: just 6 percent of male employees and 3 percent of females). Due to data limitations an accurate comparison with 2002 is not possible for different types of upper secondary vocational education.

Except for rural sub-sample, changes in returns to education between 2002 and 2005 are not statistically significant; nevertheless, there is a clear pattern: Returns to education have increased for those working in the countryside and slightly for some groups in the public sector but have decreased slightly for all other groups. This is likely due to shortage of unskilled labor caused by mass outflow of workers abroad since Latvia has joined the EU in May 2004.

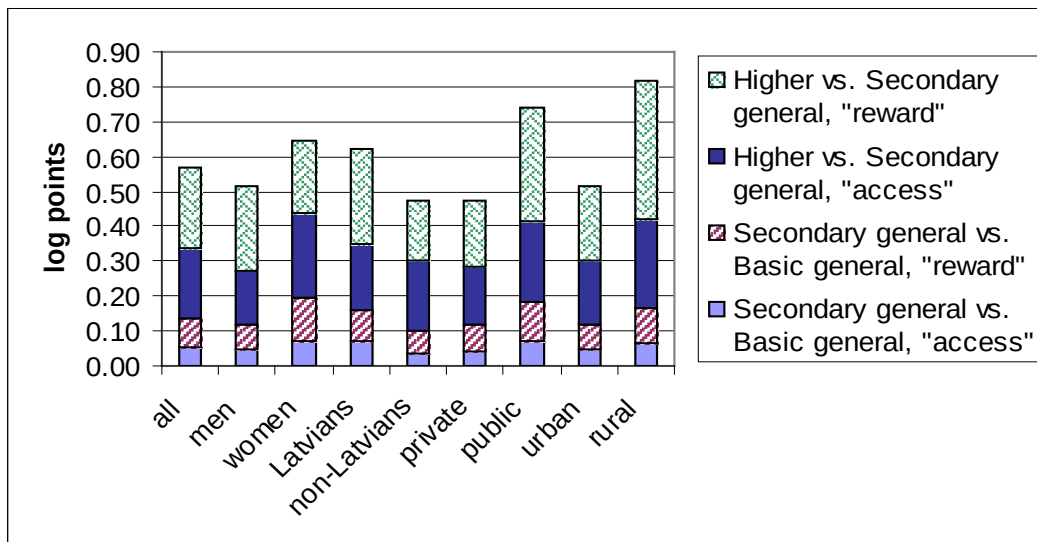
Box 1: “Access” vs. “Reward” Wage Differentials

Schooling may affect wages in several ways. More educated employees can be better paid (i.) because they are more productive, (ii.) because employers use educational attainment as a signal of abilities and/or productivity, (iii.) because they work in ‘better’ firms, or (iv.) because they occupy higher positions in firms.

Comparing earnings functions with and without controls for respondent’s occupation,¹ and firm size allows decomposition of the total payoff to education into two components: (a) returns via access to better firms and/or higher positions, which we refer to as “access” returns, and (b) wage differentials within major groups of occupations and firms, which we refer to as “reward”. The results of the decomposition of the wage differentials are presented in Figure 2.2.

On average, 46 percent of the returns to higher education (as compared to secondary) are of the “access” type. This share is larger for women than for men (54 vs. 39 percent), and for non-Latvians than for Latvians (54 vs. 41 percent). The latter fact suggests an important development (in 2002, the access component for non-Latvians was just 40 percent). As for secondary general education, the access component or returns is 41 percent on average; this share is particularly low (34 percent) among non-Latvians and private sector employees, as well as among women (37 percent).

**Figure 2.2 Returns to education in Latvian labor market
by source and group of employees, 2005.**



Notes: “Reward” component is returns to education when major group of occupations and firms size are controlled. “Access” component is difference between returns without and with occupation and firm size controls. Returns are measured in logarithmic points rather than percent to allow decomposition. Professional higher education is not separated from academic. Presented results refer to full-time employees.

Source: Calculation based on LFS data.

¹ 9 major groups according to International Standard Classification of Occupations.

B. EXPERIENCE AND EARNINGS

Age-earnings profiles. Cross-sectional age-earnings profiles describe how earnings of individuals differ across cohorts at a given point in time. Age is often taken as a proxy for the length of work experience in studies of returns to human capital. Thus age-earnings profiles give information about how experience is rewarded in the labor market. On the other hand, these profiles are related to on-the-job training (which raises workers' productivity and hence earnings). The typical industrialized country age-earnings profile for men in most education groups is rising up to about age of 50, after which it decreases. The picture is similar also for women, except college graduates; for the latter category, age-earnings profile often peaks somewhat earlier.²

Compared to the socialist era, a significant decline in returns to experience has been well-documented in many post-communist countries (see e.g. Rutkowski 1996 for Poland; Kertesi and Kollo 2001 for Hungary; Rutkowski 2001 reports very low returns to experience for Bulgaria in 1997 and Macedonia in 1996). The decline is usually attributable to the fact that skills acquired under central planning have lost value during the transition. Another (likely more important reason) is that the value of such characteristics as flexibility, adaptability, ability to learn, market-oriented education, foreign language and computer skills – all of which tend to be stronger for younger workers – has increased. As a consequence, both observed and estimated age-earnings profiles in some transition economies (including all three Baltic countries, see OECD 2003; Hazans 2005) tend to peak at a much earlier age than in other industrialized countries. In other transition countries, however, returns to experience have not declined (see Munich *et al* 2005 for Czech R.) or have declined only for younger workers (Orazem and Vodopivec 1995). Moreover, age-earnings profile peak near 50 years of age in Czech R., Slovakia, and Hungary (see Munich *et al* 2005, Jurajda 2003, and Kertesi and Kollo 2001 respectively). This might be due to institutional and/or social forces supporting the advantage of older workers, as well as because a slower pace of the reforms than in the Baltic countries.

The Latvian LFS 2005 can be used to estimate age-earnings profiles separately by gender and education. Results are summarized in Figure 2.3 (models are not presented here; they do not include marital status and tenure, which strongly correlate with age, but include (significant) interactions of age and age squared with education levels; on the other hand, slightly more detailed classification of economic activities is applied).

The shape of the male profiles is the same for all types of secondary education (including postsecondary vocational). The male earnings profile peaks at 38 years of age for workers with higher education, and at 34 otherwise, except for persons who obtained vocational education after general basic education. For the latter group, the profile is downward-sloping, indicating that quality of this type of education has improved recently.

² See e.g. Ehrenberg and Smith (2005), pp. 289 – 296. Note that the classic Mincer's (1974) specification of the earnings function uses so called *potential experience* = age – 6 – years of schooling; this approach, however is not well suited for Latvia for two reasons. First, number of years of schooling is not reported in the LFS, and different types of tertiary education, e.g. BA and MA, which may imply from 3 to 6 years of studies, are not distinguished. Second, many (although not all) tertiary students work, and for them Mincer's formula would underestimate experience.

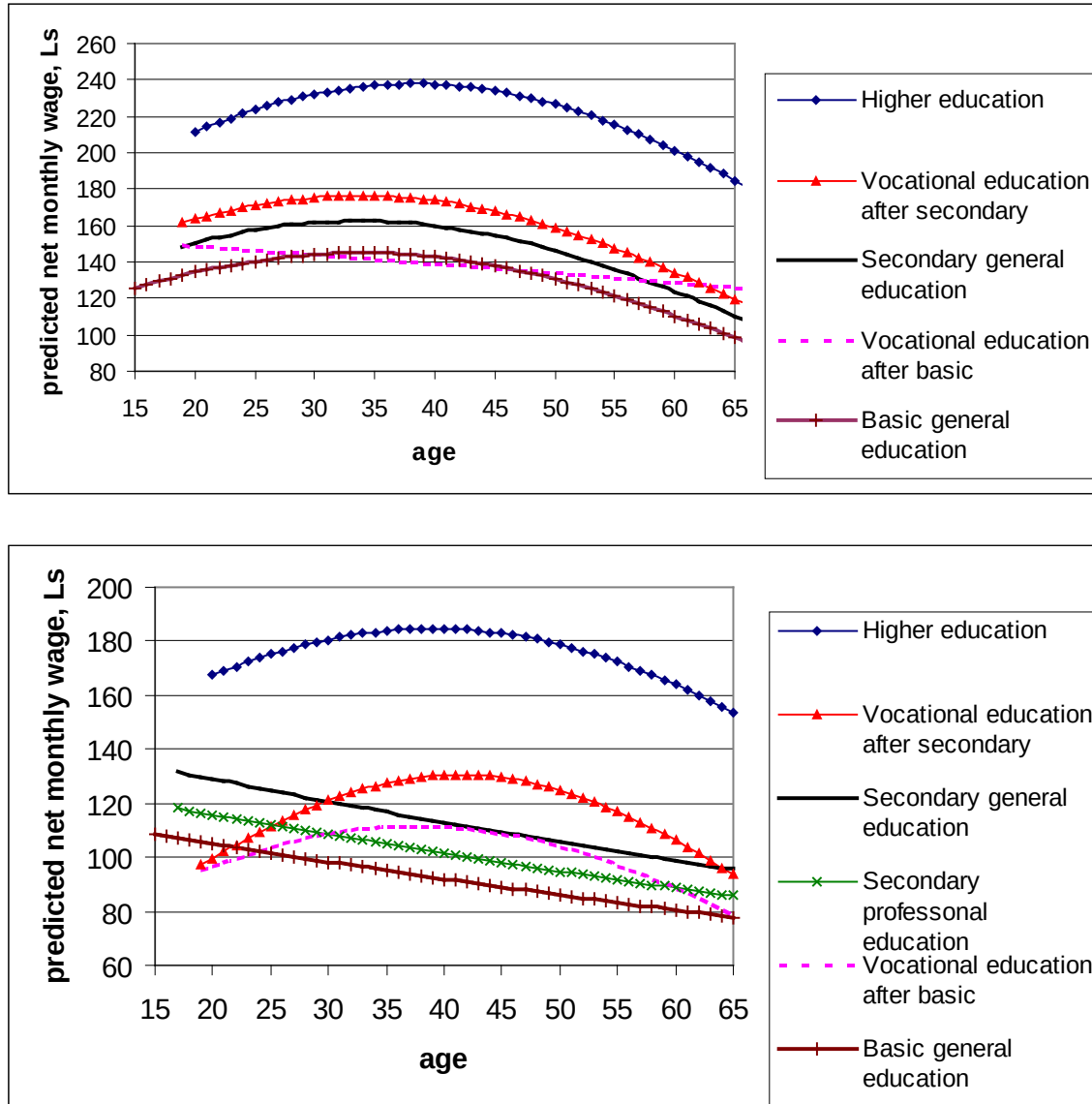
The fact that earnings of men with higher education peak later suggests that returns to experience are somewhat larger for workers with higher education. Both for higher and for secondary education, however, peak earnings exceed predicted earnings for a worker, who is by 15 years younger, by about 9 percent. This difference is remarkably small in comparison e.g. with USA, where average earnings of a 38 years old college graduate working full-time in 2003 exceeded earnings of his/her 23 years old counterpart by more than 100 percent for males and by almost 70 percent for females; for workers with secondary education, this differential exceeds 60 percent for males and 50 percent for females (Ehrenberg and Smith 2005, pp. 289 – 290).

Compared to 2002 profiles reported in Hazans (2005), the age of peak earnings is found three years later for men with higher education, but three years earlier for men with basic general education, and one year earlier for men with secondary general education. This suggests that return to experience has slightly increased for highly skilled male workers, but in other skill groups young workers have further improved their position. However, the shape of profiles has not changed significantly.

For females with higher education, like for their male counterparts, the peak of the age-earnings profile is at 38 years. This profile is just slightly flatter than for males: predicted wage at 38 years is just 7.2 percent above its value at 23 years of age, while for males the difference is 8.5 percent. Returns to experience are larger for women aged 20 to 40 with vocational education obtained after general secondary or basic education: the *ceteris paribus* wage differential between 38 years old and 23 years old female worker is 21.4 percent in the former group and 10.4 percent in the latter. However, after age of 41 (respectively, 38) earnings fall also in this skill groups.

For females with secondary general, secondary vocational or basic general education age-earnings profiles are downward sloping, indicating a strong position of young employees.

Figure 2.3 Estimated Age – Earnings Profiles of full-time employees. Latvia, 2005
(Upper panel: males, lower panel: females)



Notes: The profile for men with secondary vocational education (not shown) has the same shape as for secondary general and postsecondary vocational education and is located between them. For both genders the profile for vocational basic education is not different from that for general secondary education.

Tenure effects on earnings. It is well documented that even after controlling for age (and other factors), earnings rise with job tenure – the number of years with the current employer (see Ehrenberg and Smith (2005), pp. 380-382 and references therein). There are several competing (and, plausibly, complementary) explanations: (i) poor job matches tend to end quickly, hence longer tenures are associated with better matches and hence higher productivity; (ii) accumulation of firm-specific human capital (via learning by doing, team-work, and on-the-job training) leads to increase in worker's productivity as

her tenure gets longer; (iii) some employers apply delayed compensation schemes. If occupation (respectively, firm or plant size) is not included among controls, returns to tenure reflect also promotion possibilities (respectively, growth of the firm).

In Latvia, the average tenure of full-time employees (working at least 35 hours per week) in 2005 was 6.5 years for men and 8.5 years for women. Other things equal, an additional year of tenure increased wages on average by 1.2 percent. Of this increase, 0.7 percent represented a “reward”, in the sense that it remains after controlling for occupation and firm size. The remaining 0.5 percent can be attributed to promotions associated with longer tenure. This decomposition looks similarly also for men and women (see Table 2.3). The tenure premium, as usual, becomes smaller for longer tenures (the figures reported above are calculated at mean values). The first years of tenure commands the highest premium: 6.9 percent on average. This premium is much larger for Latvians than for non-Latvians (8.6 vs. 4.4 percent), as well as in urban areas as opposed to the countryside (8.1 vs. 2.0 percent). Moreover, the significant wage increase after the first year with the firm is captured mostly by the young: Among full-time workers, the proportion of those with tenure less than one year is 74 percent for teenagers, 39 percent for the 20-24 year olds, 19 percent for the 25-34 year olds, 13 percent for the 35-54 year olds, and 8 percent for the older ones.

Temporary workers are catching up. Another indicator of the strength of employment relationships is the type of contract. Temporary or seasonal workers earn 8.5 percent less than permanent workers with the same characteristics; this gap is somewhat smaller than three years ago; moreover, it is not significant when occupation and firm size are also controlled for. On top of this, the share of fixed-term workers among fulltime employees went down from 11.5 percent in 2002 to 6.9 percent in 2005. Plausibly, the new labor code introduced in 2002, which restricts both total duration of temporary contracts with one employer and the categories of workers which can be employed under fixed-term contracts, has played its role. Another relevant development is growing incidence of “project employment”, which usually involves high quality jobs.

Notwithstanding the above mentioned developments, temporary workers on average still are underpaid. This can be because many fixed-term workers for various reasons have low reservation wages (those who have failed to find permanent jobs, new entrants, workers with negative unobserved characteristics etc). Such workers face relatively high unemployment risks as well as low wages (see van Ommeren and Hazans 2005 for a detailed discussion and further references). Any compensating wage differentials are overwhelmed by other effects, such as unobservable differences in quality between workers with temporary and permanent jobs. On top of this, temporary workers receive less on-the-job training and are rarely promoted. Table 2.4 present shares of temporary workers and *ceteris paribus* wage gap between them and permanent workers by group of employees.

Incidence of temporary work is relatively high among men (8.7 percent), in the private sector (9.1 percent), and in the rural areas (12 percent). It is low among women (4.9 percent) and in the public sector (2.5 percent). While number of fixed-term workers in the public sector is small, this is where they are severely underpaid. The negative wage

differential for temporary workers in 2005 was larger for minority workers (-11.1 percent) than for ethnic Latvians (-7.9 percent), while it was the other way around in 2002. The difference is, however, not statistically significant; moreover, the incidence of temporary work is slightly higher among Latvians (7.2 percent against 6.4 percent for minority workers).

C. WAGE DISCRIMINATION AND SEGREGATION

Gender pay gap. Wage discrimination refers to a situation in which equally productive individuals are paid different wages due to some characteristic (like gender, or ethnicity) that is unrelated with productivity. The issue of the gender pay gap has been a focus of research and policy for both for transition and developed market economies (see e. g. Bayard *et al* (2003), Jurajda (2003; 2005), Jurajda and Harmgart (2004), and references therein). Here we apply Oaxaca – Ransom (1994) refinement of the classic Oaxaca (1973) decomposition of the gender wage gap into explained and unexplained part. The explained part of the gap (productivity differential) is the differential in (geometric) mean predicted wages of males and females using earnings function estimated over pooled sample without gender dummy. This earnings function is thus assumed to represent the “fair” wage structure. We use net monthly earnings at the main job as the wage measure.

On average, in 2005 men’s monthly wage in Latvia exceeded women’s wage by almost 25 percent, while according to observed productive characteristics the difference should be less than 5 percent. Even after accounting for main group of occupation and firm size the gender pay gap should be less than 8 percent. When two-digit occupation (rather than main group of occupation) is controlled for, the explained part of the gap increases only slightly and equals 11.4 percent. Total gender wage gap has declined since 2002 but its structure is largely unchanged: about two thirds of the gap is unexplained by observed productive characteristics and main group of occupations; even when two-digit occupation is accounted for, the unexplained gap amounts to more than a half of the total wage differential. Only one eights (12.6 percent) of the gender wage gap is due to occupational segregation within major occupational groups. Productivity differential without occupation controls absorbs just about one fifth of the gender wage gap (see Table 2.5 for details).

The **ethnic wage gap** (9.6 percent in 2005) is much smaller than the gender gap: It has declined slightly since 2002 but is almost completely unexplained by the productive characteristics observed in the LFS (see Table 2.6 for details). However, knowledge of Latvian language is an important productive characteristic which is unobserved in the LFS.

Results from a representative survey of employees in late 2005 – early 2006 suggest that when language skills are accounted for, the unexplained gap is reduced substantially (Table 2.7):

- workers who are not native Latvian speakers but have good knowledge of Latvian language receive on average 2 percent higher wages; according to productive characteristics other than language they should be paid 6 percent more than Latvians; hence the “unexplained” gap is just 4 percent (significant at 5% level), which plausibly is a fair premium for the difference between perfect and good language skills; for this group of workers occupational distribution differs very little from that of native Latvian speakers;
- the raw wage gap between native speakers and workers, who evaluate their Latvian language skills as medium, is 10 percent, while according to observed productive characteristics other than language it should be less than 4 percent; the “unexplained” gap is about 6 percent (significant at 1% level), which again can be seen as a quite modest premium for the difference in language skill level; this group of workers feature substantial occupational segregation from native speakers, plausibly occupying positions where good knowledge of state language is not so critical;
- the raw wage gap between native speakers and workers with poor knowledge of Latvian language is 13.4 percent, and it is almost completely explained by observed productive characteristics other than language; workers with poor knowledge of Latvian language are concentrated mainly in manual jobs and feature very substantial occupational segregation from native Latvian speakers.

D. REGIONAL EFFECTS.

Earnings functions may also shed light on the degree of geographical segmentation of labor markets due to barriers to mobility. The inclusion of regional variables representing the capital city, and the five statistical regions (Pierīga, Vidzeme, Kurzeme, Zemgale, un Latgale) in the earnings function generally have highly significant effects. Table 2.8 compares the regional effects in 2005 and in 2002. When more than two regions (or other groups) are involved, such a comparison is best done using *normalized regression* approach (see Yun 2005), which basically identifies deviations from [non-weighted] mean level rather than from a fixed reference group (the problem with the latter, traditional, approach is that the position of the reference group in the distribution is also changing).

Regional wages less polarized than 3 years ago. Table 2.8 documents that

- While wages in the capital city remain significantly higher than in the rest of the country, and wages in Latgale – significantly lower than in other regions, both gaps have declined since 2002 (note that development between 1997 and 2002 have been opposite, see Hazans 2005).

- The positions of Pierīga and Kurzeme regions, as well as Rīga, in the earnings rankings have worsened by 5 to 6 percentage points, while Zemgale region has made an impressive progress (almost 12 percentage points up).

Group-specific regional effects. Table 2.9 provides regional wage differentials by subpopulation. As it has been shown in Table 2.8, Vidzeme's position has not changed since 2002, hence it can be safely used as the reference group for comparisons with other regions. Results suggest that:

- the positive wage effect of working in Rīga for Latvians (35 percent) is much larger than for non-Latvians (22 percent);
- the positive wage effect of working in Zemgale for men (24 percent) is more than twice as big as for women (11 percent);
- the negative wage effect of working in Latgale is substantial (around 14 percent) and strongly significant for men, for non-Latvians, for the private sector workers, and in urban areas; it is not significant for women, in the public sector, and in the countryside, while for Latvians it is significant but not as sizable as for non-Latvians.

Rural-urban earnings gap declining. In 2005, earnings of employees working in rural areas were on average just 2.5 percent below earnings of their otherwise similar counterparts working in cities outside Rīga; this difference is not statistically significant (see Table 2.9). This is an important development since 2002, when the rural-urban wage gap was significant at 1% level and amounted to -10 percent. In 2005, similar gap is found only in the public sector. Ethnic Latvians working in rural areas earn 5 percent less than their urban counterparts (other things equal), while for non-Latvians rural-urban wage differential appears to be positive (7 percent).

When dummies for 6 main cities (Jurmala in Rīga region; Liepāja and Ventspils in Kurzeme; Jelgava in Zemgale; Daugavpils and Rēzekne in Latgale) are included in the models, it appears that, other things equal, workers in Ventspils receive 21% more than elsewhere in Kurzeme, while other effects are not significant.

A substantial part of the effect of job location on wages can be explained by differences in unemployment rates (this effect is known as “the wage curve”, see Blanchflower and Oswald 1994). When local unemployment (measured as registered unemployment rate by the 33 NUTS-4 regions in the end of year 2004) is controlled for, earnings in Kurzeme and Latgale are not significantly different from earnings of otherwise similar workers in Vidzeme (although the 7 percent difference between Kurzeme and Latgale remains significant). Moreover, advantage of Rīga over Vidzeme is reduced from 31 to 22 percent (Table 2.10). Wage curve elasticity (which measure the effect of local unemployment on wages) takes value -0.11 (very close to its classic value of -0.10, see Blanchflower and Oswald 1994). The estimated elasticity is significant at 5% level even when White (1984) robust standard errors are allowed to be arbitrary correlated within the 33 NUTS-4 regions to account for the Moulton (1990) problem.

In the private sector, wage curve elasticity is almost twice as large as on average, while public sector wages are not responsive to local unemployment rate (see (Table 2.10 for details). Like found in other studies (e. g. Morrison et al 2006), wage curve elasticity is higher for men than for women. This is at least in part related to the fact that proportion of public sector workers is larger among women (42 percent) than among men (26 percent). Another reason is that proportion of workers with tertiary education is also higher among women (28 percent) than among men (17 percent). According to the theory (Topel 1986, Johnsen 1999, Morrison et al 2006), wages of less geographically mobile workers are more likely to be affected by local labor market conditions. In Latvia, workers with tertiary education are more mobile in terms of both commuting (Hazans 2004, Table 12) and internal migration (Hazans 2003, Table 2.1). Hence, one can expect lower wage curve elasticity among these workers. Indeed, when local unemployment rate is interacted with higher education dummy, the estimated interaction is highly significant and positive, with a value 0.18 which exceeds absolute value of the main effect of the (log) unemployment (-0.14). In other words, there is no wage curve for employees with higher education.

E. INDUSTRY WAGE DIFFERENTIALS

During the last three years, there have been important changes in relative wages of workers employed in various industries (see Table 2.11 for details). Most important, education and especially health workers have improved their position (by 4 and 10 percentage points respectively); by contrast, employees in other personal services have lost 5 points. Agricultural and manufacturing workers have lost about 4 percentage points, while those employed in forestry and fishing gained 6 points.

F. CHANGES IN OVERALL EARNINGS STRUCTURE

It follows from the discussion above that the following groups have slightly improved their relative position in earnings distribution:

- females;
- ethnic minorities;
- temporary workers;
- new entrants;
- private sector workers;
- workers employed in forestry and fishing, health, education, transport and communication;
- those working in rural areas and in Zemgale region;
- persons with vocational basic education (this is a small group though).

Table 2.12 presents detailed decomposition analysis of the change in nominal net monthly earnings between 2002 and 2005 based on estimated earnings functions.

Box 2: Detailed wage decomposition. The purpose of decomposition analysis (see Oaxaca (1973), Oaxaca and Ransom (1994), Yun (2005)) is to decompose the total change into three components:

(i) *characteristics' effect* due to difference in average [observed] characteristics of workers and firms; (ii) *coefficients' effect* due to difference in returns to observed characteristics; (iii) *intercept effect* caused by unobserved factors (the intercept effect is sometimes included in the coefficients' effect).

The following stylized example helps to understand the idea of decomposition (for simplicity, wage equation is presented not in logarithmic form). Assume in period 1 there are 5 workers: two women (wage = 70) and three men (wage = 100). Normalized wage equation is $W_1 = 100M + 70F = 85 + 15M - 15F$, where M and F are the proportions of men and women respectively, and 85 is non-weighted average of male and female wage rates. In period 2, one of the men emigrates, and his job is taken by a woman. New wage rates are 120 and 100 (instead of 100 and 70), so that wage equation takes form $W_2 = 120M + 100F = 110 + 10M - 10F$. Total change in wage $E[W_2] - E[W_1] = 108 - 88 = 20$ may be decomposed as follows: (i) characteristics' effect $= 15(0.4 - 0.6) - 15(0.6 - 0.4) = -6$ is due to change in composition of the workforce in favor of low paid workers – women; (ii) coefficients' effect $= (10 - 15)0.4 + (-10 + 15)0.6 = 1$ is positive; this is because proportion of women (whose relative position has improved) in period 2 is higher than proportion of men; (iii) intercept effect $= 110 - 85 = 25$ reflects wage increase not related to gender (the only observed characteristic in the model).

Overall, the model predicts nominal wage growth of 31.3 percent ($\exp(0.272)=1.313$) between 2002 and 2005. About one fifth of this growth is explained by increase in returns to hours worked, and another 8 percent (i.e., 2.3 percentage points of wage increase) are explained by reallocation between sectors, increased average job tenure, as well as (to a smaller extent) changing returns to these characteristics. On the other hand, substantial negative contributions to wage change are due to changes in returns to age (-28 percent of the total increase), education (-10 percent) and job location (-15 percent). This means that workers whose position in wage distribution has improved (e.g. young, low-skilled, or rural) belong to relatively small groups. Contribution of change in mean values of these characteristics were also negative but small. Contribution of change in other average observed characteristics of workers (ethnicity, gender, contract type) and returns to these characteristics is almost negligible. This is despite improved *relative* position of disadvantaged groups (ethnic minorities, females, fixed-term workers)

As the result, 134.6% of the whole wage increase has to be attributed to unobserved (or not included in the model) factors, i.e. to the intercept. Put another way, if the distribution of full-time workers with respect to education, age, gender, ethnicity, marital status, contract type, hours worked, tenure, sector of economic activity, and job location, as well as returns to these characteristics would not change, the nominal wage would increase by 44.3 percent ($\exp(0.367)=1.443$). This may include productivity growth, better access to foreign markets (and foreign customers in Latvia), upward pressure on wages due to outflow of labor to other EU countries, as well as due to non-wage inflation. Of course increase of minimum wage by one third (from Ls 60 in 2002 to Ls 80 in 2005), as well as increase in the income tax main deductible from Ls 21 in 2002 to Ls 26 in 2005, and increase in the deductible per supportable family member from Ls10.50 in 2002 to Ls 18 in 2005 have played an important role. It is worth noticing that about a half of the predicted wage growth has been eliminated by inflation (16.7 percent between 2002 and 2005).

CONCLUDING REMARKS

In 2002-2005, Latvian labor market has witnessed dramatic changes related both to unprecedented economic growth (28 percent in 3 years) and to massive outflow of labor force after EU enlargement in May 2004. Both in 2000-2005 and in 2002-2005, Latvian employment rate has been rising faster than that of any other new EU member state and also faster than that of any other EU member except Spain. Expanding economic opportunities in the labor market appear to have been the primary channel through which the benefits of rapid economic growth were shared widely across the population during this period.

Emerging shortage of labor has led to strong growth of real wages and reduction of unemployment. Moreover, it has increased employment rates and improved labor market position of ethnic minorities, elderly, fixed-term workers, the low-skilled, and other disadvantaged groups. Wage growth, in turn, has increased labor force participation, resulting in further increase of employment rates. This study, based on micro-level analysis of Latvian Labor Force surveys 2002-2005, has provided a detailed analysis of these changes, with a particular focus on developments which are likely to improve living standards of formerly disadvantaged segments of population.

Unemployment risk has become spread much more evenly across social groups. Unemployment rates have declined in all age groups for men and in most age groups for women; for both genders the largest reduction of unemployment occurred for 35 to 44 year olds, and for the youth. Youth unemployment is not a serious problem anymore. Age-unemployment risk profile has become flat for men aged 20 to 74 (with a very modest increase between 45 and 54 years of age). For women, unemployment risk is declining up to 34 years of age (other things equal) and almost flat thereafter.

Latvia's regions have become less polarized in terms of both unemployment and earnings. The largest reductions of unemployment risk have occurred for men (especially those with less than upper secondary education) in Zemgale, Kurzeme, and Latgale regions. Nevertheless, even in 2005 predicted unemployment risk remained high for low-skilled men and women in Latgale, Vidzeme, and Kurzeme. While wages in the capital city remain significantly higher than in the rest of the country, and wages in Latgale region – significantly lower than in other regions, both gaps have declined since 2002. Similarly, the rural-urban earnings gap has declined considerably in 2005 relative to 2002 - in fact, in 2005, average earnings of employees working in rural areas were statistically indistinguishable from those of otherwise similar counterparts working in urban areas. Plausibly, both improved internal mobility of labor and external mobility shock have contributed to these developments.

While minority population still has somewhat lower employment rate, by 2005 the overall gap has been reduced from more than 6 percentage points in 2002 to less than 3 percentage points in 2005. Moreover, the increase in employment between 2002 and 2005 has taken place primarily (and as far as women are concerned, exclusively) within minority population. For men, the *ceteris paribus* ethnic gaps in participation and employment have both disappeared. The ethnic gap in employment rates has narrowed

down in all age groups except the youth and the elderly. Nevertheless, the ethnic gap in employment remains substantial (10 percentage points) for persons with tertiary education. While this is just half of the gap width in 1997), there have been no significant change in this respect since 2002.

The ethnic wage gap (9.6 percent in 2005 according to LFS) is almost completely unexplained by education, age, occupational characteristics or similar variables observed in the LFS. There is also a substantial difference in distribution of ethnic Latvians and minority employees among occupations and sectors of economic activity: Latvians are over-represented in highly skilled non-manual occupations, in non-market services and agriculture, and in public sector. On the other hand, non-Latvians are found more frequently in skilled manual and elementary occupations, in industry and market services, and in private sector.

However, analysis of data from a recent nationally representative survey of employees in Latvia suggests that the difference between the two ethnic groups stem from differences in language skills. Among employees with good Latvian language skills just 7 percent would have to change occupation to make their occupational distribution identical to that of native Latvian speakers', while for workers with medium and poor Latvian language skills this proportion is, respectively, 25 and 49 percent. Similarly, once differences in language skills between the two groups are accounted for, the "unexplained" gap in earnings between the two groups is also substantially reduced.

By 2005, like three years ago, returns to tertiary education in Latvia are high international standards, while returns to secondary education are low. Returns to experience are extraordinarily low. There is some evidence that shortage of unskilled labor has led to a slight wage compression across education groups, as well as to further improvement in position of young workers in wage distribution among low- and medium-skilled.

The findings of this study suggest that labor market flexibility in Latvia improved considerably between 2002 and 2005. Moreover, responding to strong economic growth combined with external labor mobility shock, Latvian labor market in just three years has featured changes size of which is well beyond expectations.

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Table 1.1: Employment rates (%) by ethnicity and demographic group, 2002-2005.**Population aged 15-74**

	Latvians				Non-Latvians				Gap			
	2002	2003	2004	2005	2002	2003	2004	2005	2002	2003	2004	2005
Education:												
Less than secondary	30.3	28.9	27.5	28.8	24.1	25.1	24.8	27.8	6.2	3.8	2.7	1.0
Upper secondary	65.1	66.3	65.1	66.0	58.9	61.1	62.6	62.2	6.2	5.2	2.5	3.8
Tertiary	79.5	81.5	81.6	82.8	69.2	68.4	73.1	72.4	10.3	13.1	8.5	10.4
Gender:												
Men	61.3	62.4	62.2	63.3	57.1	58.8	60.3	62.3	4.2	3.7	1.9	1.0
Women	53.3	53.3	52.6	53.8	45.6	46.8	49.5	49.4	7.7	6.5	3.1	4.4
Age:												
15-24	31.2	31.6	30.0	33.7	30.5	31.2	31.6	30.1	0.7	0.4	-1.6	3.6
25-34	80.0	80.0	77.5	76.6	68.0	71.0	72.8	74.1	12.1	9.0	4.7	2.5
35-44	80.4	84.1	83.9	83.6	71.8	75.0	75.8	79.9	8.5	9.1	8.1	3.7
45-54	81.1	80.0	79.4	80.1	72.5	70.6	73.9	74.5	8.6	9.4	5.5	5.6
55-64	48.1	46.3	48.8	51.5	34.1	41.1	46.8	47.4	13.9	5.2	2.0	4.0
65-74	14.8	14.3	15.7	17.0	7.6	6.8	9.1	10.0	7.2	7.5	6.6	7.0
Overall ^a	57.1	57.6	57.1	58.2	50.8	52.2	54.5	55.4	6.3	5.4	2.7	2.8

Note: ^a Overall denotes population 15-74 years, and includes also persons with unknown level of education.

Source: Calculation based on LFS data.

Table 1.2: Employment rates (%) by ethnicity and residence, 2002-2005.**Population aged 15-74**

	Latvians				Non-Latvians				Gap			
	2002	2003	2004	2005	2002	2003	2004	2005	2002	2003	2004	2005
Settlement												
Capital (Riga)	61.4	64.3	63.3	65.2	55.9	56.4	59.5	60.1	5.5	7.9	3.9	5.1
Big cities ^a	53.0	56.1	54.8	54.7	45.7	47.2	49.3	50.9	7.3	8.9	5.5	3.7
Small cities	56.3	56.5	55.6	58.3	47.7	50.6	54.0	53.8	8.6	5.9	1.5	4.6
Rural	56.1	54.7	54.7	55.0	47.4	50.0	50.4	51.2	8.7	4.7	4.3	3.8
Overall ^a	57.1	57.6	57.1	58.2	50.8	52.2	54.5	55.4	6.3	5.4	2.7	2.8

Note: ^a Daugavpils, Jurmala, Jelgava, Liepaja, Rēzekne, Ventspils.

Source: Calculation based on LFS data

**Table 1.3: Determinants of employment.
Population aged 15-74, 2002 and 2005**

	<i>Survey probit estimates</i>							
	men		men		women		women	
	2002 (P=0.596)		2005 (P=0.629)		2002 (P=0.499)		2005 (P=0.521)	
	dP/dx	t-value ^a	dP/dx	t-value ^a	dP/dx	t-value ^a	dP/dx	t-value ^a
Education								
(vs. basic or less)								
higher	0.223	10.39***	0.233	10.56***	0.272	14.05***	0.299	13.67***
general secondary	0.117	6.70***	0.089	4.95***	0.098	5.57***	0.098	4.98***
postsecondary or upper secondary								
vocational ^b	0.153	9.14***	0.140	8.13***	0.164	9.6***	0.182	9.27***
basic with vocational	0.148	3.37***	0.169	4.07***	0.060	1.16	0.090	1.56
Non-Latvian	-0.053	-4.47***	-0.006	-0.53	-0.092	-8.33***	-0.059	-4.81***
Single	-0.146	-8.86***	-0.153	-9.48***	-0.007	-0.67	-0.006	-0.49
With children	0.058	3.34***	0.051	2.82***	-0.036	-2.64***	-0.077	-4.89***
Age (vs. 35-44)								
15-19	-0.375	-12.52***	-0.455	-14.38***	-0.623	-21.13***	-0.629	-17.87***
20-24	0.012	0.49	-0.021	-0.92	-0.270	-10.44***	-0.314	-11.52***
25-34	0.100	5.35***	0.040	2.09**	-0.064	-3.28***	-0.085	-4.2***
45-54	0.020	1.00	-0.027	-1.32	0.010	0.56	-0.061	-3.04***
55-64	-0.208	-8.44***	-0.254	-9.82***	-0.376	-16.06***	-0.357	-14.46***
65-74	-0.510	-17.14***	-0.558	-18.58***	-0.633	-26.07***	-0.672	-25.06***
Residence ^c								
(vs. Vidzeme and Urban)								
Riga region	0.020	0.99	0.035	1.78*	0.002	0.13	0.020	1.1
Zemgale	-0.014	-0.58	0.008	0.35	-0.020	-0.94	-0.004	-0.17
Kurzeme	-0.044	-1.83*	0.030	1.31	-0.049	-2.14**	-0.009	-0.41
Latgale	-0.121	-5.19***	-0.109	-4.75***	-0.047	-2.23**	-0.052	-2.52**
Rural	0.032	2.48**	0.022	1.62	0.034	2.79***	0.002	0.13
Number obs.	8701		8148		10474		9521	

Note: ^a t-values are based on robust standard errors adjusted for clustering on households. Columns dP/dx report means of individual marginal effects (change in predicted probability of participation for given category compared to reference category, other things equal). ^b Consistent distinguishing between different types of vocational education was problematic, hence they were merged in a single category. ^c Definition of regions has been changed in 2004. The new definition is applied here both for 2002 and for 2005.

*, **, *** - estimates significantly different from zero at 10%, 5%, 1% level respectively.

Source: Calculations based on LFS data.

**Table 1.4: Determinants of labor force participation.
Population aged 15-74, 2002 and 2005**

	<i>Survey probit estimates</i>							
	men		men		women		women	
	2002 (P=0.684)		2005 (P=0.691)		2002 (P=0.560)		2005 (P=0.569)	
	dP/dx	t-value ^a	dP/dx	t-value ^a	dP/dx	t-value ^a	dP/dx	t-value ^a
Education								
(vs. basic or less)								
higher	0.139	7.34***	0.190	9.08***	0.244	11.7***	0.266	12.7***
general secondary	0.070	4.43***	0.064	3.78***	0.106	6.09***	0.089	4.91***
postsecondary or upper secondary								
vocational ^b	0.109	6.97***	0.129	8.01***	0.167	9.68***	0.168	8.98***
basic with vocational	0.102	2.79***	0.132	3.61***	0.080	1.62	0.075	1.28
Non-Latvian	-0.031	-2.94***	0.002	0.17	-0.072	-6.51***	-0.035	-2.95***
Single	-0.102	-6.65***	-0.132	-9.00***	0.001	0.08	0.003	0.27
With children	0.063	3.78***	0.060	3.19***	-0.047	-3.50***	-0.074	-4.73***
Age (vs. 35-44)								
15-19	-0.474	-16.0***	-0.457	-14.37***	-0.656	-21.6***	-0.658	-18.76***
20-24	-0.046	-2.16**	-0.049	-2.37**	-0.256	-11.3***	-0.309	-11.51***
25-34	0.065	4.22***	0.037	2.2**	-0.070	-4.19***	-0.066	-3.53***
45-54	-0.011	-0.69	-0.013	-0.7	-0.015	-0.94	-0.042	-2.22**
55-64	-0.287	-12.7***	-0.272	-10.9***	-0.447	-18.6***	-0.388	-16.64***
65-74	-0.642	-20.1***	-0.623	-20.48***	-0.724	-26.6***	-0.727	-26.26***
Residence^c								
(vs. Vidzeme and Urban)								
Riga region	0.015	0.73	0.016	0.86	-0.011	-0.52	0.019	1.05
Zemgale	-0.017	-0.73	-0.031	-1.4	-0.035	-1.45	-0.010	-0.44
Kurzeme	-0.036	-1.71*	0.023	0.99	-0.056	-2.19**	0.012	0.54
Latgale	-0.096	-3.58***	-0.097	-4.27***	-0.042	-1.6	-0.052	-2.44**
Rural	-0.002	-0.13	0.001	0.11	-0.003	-0.19	-0.015	-1.14
Number obs.	8701		8148		10474		9521	

Note: ^a t-values are based on robust standard errors adjusted for clustering on primary sampling units. Columns dP/dx report means of individual marginal effects (change in predicted probability of participation for given category compared to reference category, other things equal). ^b Consistent distinguishing between different types of vocational education was problematic, hence they were merged in a single category. ^c Definition of regions has been changed in 2004. The new definition is applied here both for 2002 and for 2005. *, **, *** - estimates significantly different from zero at 10%, 5%, 1% level respectively.
Source: Calculations based on LFS data.

**Table 1.5: Labor force participation rates (%) by ethnicity and demographic group.
Population aged 15-74, 2002-2005.**

	Latvians				Non-Latvians				Gap			
	2002	2003	2004	2005	2002	2003	2004	2005	2002	2003	2004	2005
Education:												
Less than secondary	36.3	33.4	32.7	33.1	31.5	32.4	30.6	33.5	4.8	1.0	2.1	-0.4
Upper secondary	72.0	72.4	71.2	71.7	69.9	71.3	73.3	69.6	2.1	1.1	-2.1	2.0
Tertiary	83.3	84.0	83.9	84.8	74.3	74.5	78.5	78.4	8.9	9.5	5.4	6.4
Gender:												
Men	68.8	68.3	68.3	68.8	67.8	68.1	69.4	69.6	1.0	0.2	-1.1	-0.8
Women	58.2	57.6	57.0	57.6	53.3	55.0	57.6	55.7	4.9	2.6	-0.6	1.9
Age:												
15-24	38.1	36.9	35.9	38.3	40.6	41.7	40.1	35.5	-2.4	-4.8	-4.2	2.8
25-34	87.8	86.4	84.8	83.0	79.6	81.2	84.6	83.5	8.2	5.2	0.2	-0.4
35-44	88.9	90.6	89.9	88.9	84.9	86.3	85.9	87.0	4.0	4.3	4.0	1.8
45-54	88.4	87.3	85.6	86.2	81.9	82.4	85.7	84.3	6.5	4.9	-0.1	1.9
55-64	51.4	49.2	52.2	54.8	40.1	46.1	52.5	53.3	11.3	3.0	-0.4	1.5
65-74	15.6	14.7	16.0	17.6	8.8	7.4	10.1	11.1	6.7	7.3	5.9	6.6
Total	63.2	62.7	62.3	62.8	59.9	61.0	63.0	62.2	3.3	1.7	-0.7	0.6

Source: LFS data and calculation.

**Table 1.6: Population aged 15-24
by participation in education and in the labor force, 2002 and 2005**

Status	Percent					
	All		Latvians		Non-Latvians	
	2002	2005	2002	2005	2002	2005
Student, economically active	9.8	9.0	10.7	9.8	8.1	7.4
Student, economically inactive	53.2	55.8	54.7	55.8	50.5	55.7
Non-student, economically active	29.2	28.4	27.4	28.5	32.4	28.2
Non-student, economically inactive	7.8	6.8	7.1	5.9	9.0	8.8
Total	100.0	100.0	100.0	100.0	100.0	100.0
Students, discouraged	6.6	4.0	5.8	4.2	8.0	3.5
Non-students, discouraged	2.5	2.1	2.4	1.5	2.7	3.5
Discouraged, total	9.1	6.1	8.2	5.7	10.7	6.9
Economically active, total	39.0	37.4	38.1	38.3	40.6	35.5
of which						
Employed students	6.9	6.8	7.8	7.6	5.4	5.1
Employed non-students	24.0	25.8	23.4	26.2	25.1	25.0
Unemployed students	2.9	2.2	2.9	2.2	2.7	2.3
Unemployed non-students	5.2	2.6	4.1	2.4	7.4	3.1
Employed, total	30.9	32.6	31.2	33.7	30.5	30.1
Unemployed, total	8.1	4.8	7.0	4.6	10.1	5.4
Students, total	63.0	64.8	65.5	65.6	58.6	63.1
of which:						
Primary level students	11.2	10.6	11.3	11.8	11.1	7.9
Upper secondary level students	30.4	31.6	31.6	31.5	28.2	31.8
Tertiary level students	21.4	22.7	22.6	22.3	19.3	23.3
Number obs.	3833	3506	2480	2397	1353	1109

Notes: "Students" are all respondents who have defined their activity status during the *three months* before the survey as "Working student (incl. paid apprenticeship)" or "Non-working student (incl. unpaid apprenticeship)", where studies may refer to general or vocational school, college, or university. Economic activity, employment and unemployment, by contrast, are defined with respect to the survey week.

Source: Calculations based on LFS data.

**Table 1.7: Determinants of unemployment.
Population aged 15-74, 2002 and 2005**

	<i>Survey probit estimates</i>							
	men		men		women		women	
	2002 (P =0.129)		2005 (P=0.090)		2002 (P =0.110)		2005 (P=0.084)	
	dP/dx	t-value	dP/dx	t-value	dP/dx	t-value	dP/dx	t-value
Education								
(vs. basic or less)								
higher	-0.155	-7.53***	-0.094	-5.11***	-0.101	-5.39***	-0.102	-5.44***
general secondary	-0.095	-5.13***	-0.051	-2.94***	-0.022	-1.00	-0.039	-1.79*
postsecondary or upper secondary								
vocational ^b	-0.104	-5.88***	-0.049	-3.07***	-0.046	-2.28**	-0.063	-3.08***
basic with vocational	-0.104	-2.50**	-0.085	-2.32**	0.003	0.05	-0.045	-0.78
Non-Latvian	0.040	3.54***	0.012	1.20	0.049	4.66***	0.045	4.16***
Single	0.093	6.24***	0.056	4.19***	0.015	1.46	0.013	1.26
With children	-0.015	-1.02	-0.009	-0.64	0.000	-0.01	0.022	1.71*
Age (vs. 35-44)								
15-19	-0.017	-0.58	0.073	2.40**	0.264	5.98***	0.107	3.24***
20-24	-0.065	-3.32***	-0.030	-2.00**	0.080	3.67***	0.061	3.31***
25-34	-0.055	-3.62***	-0.009	-0.67	0.003	0.19	0.029	2.22**
45-54	-0.032	-1.89*	0.018	1.17	-0.025	-1.99**	0.027	1.97**
55-64	-0.041	-1.87*	0.024	1.17	-0.028	-1.60	0.004	0.27
65-74	-0.077	-1.92*	-0.015	-0.47	-0.017	-0.53	0.003	0.10
Residence^c								
Riga region	-0.015	-0.85	-0.029	-1.75*	-0.028	-1.64	-0.006	-0.35
Zemgale	-0.003	-0.13	-0.053	-2.86***	-0.020	-0.96	-0.008	-0.4
Kurzeme	0.012	0.54	-0.016	-0.81	-0.001	-0.07	0.039	1.76*
Latgale	0.065	2.74***	0.041	1.69*	0.019	0.90	0.012	0.62
Rural	-0.048	-4.10***	-0.028	-2.43**	-0.062	-5.69***	-0.029	-2.51**
Number obs.	5706		5372		5707		5303	

Note: ^a t-values are based on robust standard errors adjusted for clustering on households. Columns dP/dx report means of individual marginal effects (change in predicted probability of participation for given category compared to reference category, other things equal). ^b Consistent distinguishing between different types of vocational education was problematic, hence they were merged in a single category. ^c Definition of regions has been changed in 2004. The new definition is applied here both for 2002 and for 2005. *, **, *** - estimates significantly different from zero at 10%, 5%, 1% level respectively. Source: Calculations based on LFS data.

**Table 1.8: Unemployment rates (%) by ethnicity and demographic group.
Population aged 15-74, 2002-2005.**

	Latvians				Non-Latvians				Gap			
	2002	2003	2004	2005	2002	2003	2004	2005	2002	2003	2004	2005
Education:												
Less than secondary	16.4	13.6	15.8	12.9	23.6	22.5	19.1	17.0	7.2	8.9	3.3	4.1
Upper secondary	9.6	8.3	8.6	7.9	15.7	14.2	14.6	10.7	6.1	5.9	6.1	2.8
Tertiary	4.5	2.9	2.7	2.3	6.9	8.1	6.9	7.6	2.4	5.2	4.2	5.3
Gender:												
Men	10.9	8.6	8.9	8.0	15.8	13.8	13.1	10.5	4.8	5.1	4.2	2.5
Women	8.4	7.5	7.8	6.6	14.5	14.9	14.1	11.4	6.1	7.4	6.3	4.7
Age:												
15-24	18.3	14.4	16.6	12.0	24.9	25.3	21.2	15.2	6.6	10.9	4.6	3.3
25-34	8.9	7.4	8.7	7.8	14.6	12.5	13.9	11.3	5.8	5.1	5.2	3.5
35-44	9.6	7.2	6.7	5.9	15.4	13.1	11.8	8.2	5.8	5.9	5.1	2.3
45-54	8.3	8.3	7.2	7.0	11.5	14.2	13.7	11.6	3.2	5.9	6.5	4.6
55-64	6.5	5.8	6.5	6.0	14.9	10.9	10.9	11.0	8.4	5.0	4.4	5.0
65-74	4.9	2.8	1.8	3.8	14.2	7.7	10.1	9.8	9.2	4.9	8.2	6.0
Total	9.7	8.1	8.4	7.4	15.2	14.3	13.6	10.9	5.4	6.3	5.2	3.5

Source: LFS data and calculation.

Table 1.9: Employment status, incidence of part-time and temporary work, and wage distribution in the main job by ethnicity, 2002-2005.

Employed population (or employees) aged 15-74

	Latvians				Non-Latvians			
	2002	2003	2004	2005	2002	2003	2004	2005
Employment status								
Employee	84.2	84.7	84.7	86.9	89.3	90.7	90.9	90.7
Employer	3.0	3.1	3.6	3.4	3.5	3.4	3.1	3.5
Self-employed	7.5	7.6	7.7	7.0	4.3	3.7	3.6	3.9
Unpaid worker in a family farm or business	5.3	4.6	4.0	2.6	2.9	2.3	2.3	1.8
Total	100	100	100	100	100	100	100	100
Working time								
Full-time	86.5	89.0	88.8	91.0	88.2	91.0	91.0	92.6
Part-time, ≥ 40 (or irregular) hours per week ^a	4.1	2.4	2.1	1.1	4.1	2.0	1.4	0.7
Part-time, 15-39 hours per week	7.7	7.4	7.7	7.0	6.9	6.1	6.5	6.1
Part-time, < 15 hours per week	1.6	1.1	1.4	0.9	0.8	0.9	1.1	0.6
Total	100	100	100	100	100	100	100	100
Total part-time (as declared by workers)	13.5	11.0	11.2	9.1	11.8	9.0	9.0	7.4
Involuntary part-time	4.5	4.0	4.3	3.1	4.1	4.1	4.0	2.9
Under-employed ^b	5.0	4.1	4.4	3.9	4.6	4.7	4.7	3.4
Contract type (employees)								
Permanent	86.5	89.5	90.8	91.2	85.3	87.6	90.2	91.6
Temporary, contract for more than 6 months	8.0	5.7	5.0	4.6	9.0	6.4	5.3	3.8
Temporary, contract for 6 months or less	5.5	4.8	4.2	4.2	5.7	6.0	4.5	4.6
Total	100	100	100	100	100	100	100	100
Net monthly wage (employees)								
Corresponds to net ^c minimum wage or less	20.1	12.7	16.5	12.7	22.7	13.9	15.4	11.9
Between net ^c minimum wage and Ls 100	28.2	34.4	22.1	19.8	32.4	40.0	29.1	24.9
Between Ls 100 and Ls 200	39.9	38.5	37.1	39.2	35.8	36.2	39.9	42.4
More than Ls 200	11.0	11.8	18.1	21.6	7.9	8.0	11.8	15.7
Refused to answer	0.8	2.7	6.2	6.6	1.1	2.0	3.7	5.1
Total	100	100	100	100	100	100	100	100

Notes: ^a In Latvian statistics, this category is not considered as part-time. ^b Part-time workers, who usually work less than 40 hours per week and would like to work more if their earnings are adjusted accordingly. ^c For 2002, gross; hence, in these rows 2002 results are not comparable with other years.

Source: Calculation based on LFS data

Table 1.10: Occupation in the main job by ethnicity, 2002 and 2005.
Employees aged 15-74

Occupation in the main job	Latvians				Non-Latvians			
	2002		2005		2002		2005	
	Total	Total	Men	Women	Total	Total	Men	Women
Senior officials and managers	9.4	8.6	8.6	8.6	5.3	5.7	6.7	4.7
Professionals	14.2	13.9	9.3	18.5	10.7	9.8	6.2	13.6
Technicians; associated professionals	16.0	16.1	11.2	20.9	13.2	11.9	9.0	15.0
Total highly skilled non-manual	39.5	38.6	29.1	47.9	29.3	27.4	21.9	33.3
Clerks	5.9	6.2	3.4	8.9	4.9	7.2	2.5	12.3
Service, shop and market workers	14.0	15.8	8.8	22.8	15.8	15.5	6.9	24.7
Total low skilled non-manual	19.9	22.0	12.2	31.7	20.6	22.7	9.4	37.0
Skilled agricultural and fishery workers	2.4	1.9	2.6	1.1	1.1	1.0	0.9	1.1
Craft and related trades workers	14.5	14.3	23.7	5.0	18.5	22.0	35.6	7.4
Plant and machine operators and assemblers	10.8	11.4	19.7	3.3	14.5	12.5	18.9	5.7
Total skilled manual	27.8	27.5	46.0	9.5	34.1	35.6	55.4	14.2
Elementary occupations	12.9	11.8	12.7	10.9	16.0	14.4	13.2	15.5
Total	100	100	100	100	100	100	100	100
Dissimilarity index^a (%) between Latvians and non-Latvians:								
by four "super-groups" of occupations	10.2	11.2	9.9	14.7	10.2	11.2	9.9	14.7
by nine major groups of occupations	12.5	12.4	12.0	14.8	12.5	12.4	12.0	14.8
by 27 two-digit groups of occupations	16.7	16.5	17.5	18.7	16.7	16.5	17.5	18.7

Notes. ^a Dissimilarity index *DI* (known also as *Duncan index*, see Duncan and Duncan 1955) is a number between 0 and 1, with 0 indicating equal distribution of ethnic (or other) groups among occupations, and 1 (or 100%) indicating complete segregation. In the given context, *DI* shows the minimal proportion of non-Latvians which would have to change occupations in order to make their occupational distribution identical to that of Latvians'.

Source: Calculation based on LFS data

Table 1.11: Occupation in the main job by Latvian language skills, 2005.
Employees aged 18-64

Occupation in the main job	Latvian language skills				Total
	Native	Good	Medium	Poor	
1: Senior officials and managers	6.3	5.3	3.3	0.9	5.4
2: Professionals	15.4	13.2	6.4	2.3	13.2
3: Technicians; associated professionals	17.2	17.4	10.5	6.0	15.8
Total highly skilled non-manual (1+2+3)	38.9	35.9	20.1	9.2	34.3
4: Clerks	8.2	8.4	4.1	1.8	7.4
5: Service, shop and market workers	18.4	24.2	17.0	5.5	18.7
Total low skilled non-manual (4+5)	26.6	32.6	21.1	7.3	26.1
6: Skilled agricultural and fishery workers	1.3	0.1	0.8	3.2	1.1
7: Craft and related trades workers	11.5	12.3	21.6	28.4	13.8
8: Plant and machine operators and assemblers	11.5	9.5	19.9	21.6	12.8
Total skilled manual (6+7+8)	24.4	21.9	42.3	53.2	27.7
9 : Elementary occupations (9)	10.2	9.6	16.4	30.3	11.9
Total	100.0	100.0	100.0	100.0	100.0
Dissimilarity index^a between given category and native Latvian speakers:					
by nine major groups of occupations, %	0.0	7.0	24.7	49.0	
by four “ super-groups ” of occupations, %	0.0	5.9	24.2	49.0	
Number of observations	2515	811	487	218	4031
<i>Notes.</i> ^a Dissimilarity index <i>DI</i> (known also as <i>Duncan index</i> , see Duncan and Duncan 1955) is a number between 0 and 1, with 0 indicating equal distribution of ethnic (or other) groups among occupations, and 1 (or 100%) indicating complete segregation. In the given context, <i>DI</i> shows the minimal proportion of workers with good, medium and poor Latvian language skills which would have to change occupations in order to make their occupational distribution identical to that of native Latvian speakers.					
<i>Source:</i> Calculation based on a representative survey of employees implemented in November 2005-January 2006 by <i>RS Group</i> in the framework of the study of determinants of wages commissioned by the Ministry of Welfare for the National Program of Labor Market Studies.					

Table 1.12: Sector of economic activity in the main job by ethnicity, 2002 and 2005.
Employees aged 15-74

	Latvians				Non-Latvians			
	2002		2005		2002		2005	
	Total	Total	Men	Women	Total	Total	Men	Women
Agriculture, forestry and fishing (A+B)	9.3	8.8	13.6	4.1	5.1	4.0	5.6	2.2
Mining (C)	0.3	0.3	0.4	0.2	0.4	0.1	0.2	0.0
Manufacturing (D)	15.9	13.7	16.3	11.0	22.9	19.7	20.4	18.9
Energy, gas and water supply (E)	2.6	2.7	4.0	1.4	2.6	2.3	2.9	1.6
Construction (F)	6.0	7.3	13.4	1.3	7.2	12.0	20.8	2.6
Trade (G)	14.5	14.9	11.0	18.8	16.5	16.8	11.2	22.9
Hotels and restaurants (H)	2.4	3.0	1.6	4.3	3.4	2.6	0.9	4.3
Transport, storage and communication (I)	7.4	7.4	10.5	4.4	12.9	13.5	17.5	9.2
Financial intermediation (J)	1.8	2.2	1.5	3.0	1.0	2.0	1.7	2.3
Real estate and business activities (K)	3.4	4.0	4.3	3.7	4.9	6.1	5.3	6.9
Public administration (L)	10.3	11.4	11.9	11.0	4.4	4.8	6.3	3.3
Education (M)	12.3	11.8	4.4	19.0	7.3	6.7	2.6	11.0
Health (N)	7.1	6.5	2.0	10.9	6.4	4.9	1.3	8.9
Other services (O+P+Q)	6.6	6.0	5.1	6.9	5.0	4.6	3.4	6.0
Total	100	100	100	100	100	100	100	100
<i>Dissimilarity index</i> ^a between Latvians and non-Latvians by <i>fourteen sectors</i> of economic activities, %	18.2	20.7	19.8	21.4	18.2	20.7	19.8	21.4
<i>Agriculture, forestry and fishing</i> (A+B)	9.3	8.8	13.6	4.1	5.1	4.0	5.6	2.2
<i>Industry</i> (C+D+E+F)	24.9	23.9	34.1	13.9	33.1	34.0	44.3	23.0
<i>Market services</i> (G+H+I+J+K)	29.5	31.6	28.9	34.2	38.7	40.9	36.6	45.6
<i>Non-market services</i> (L+M+N+O+P+Q)	36.3	35.7	23.4	47.9	23.2	21.1	13.6	29.2
<i>Dissimilarity index</i> ^a between Latvians and non-Latvians by <i>four main sectors</i> of economic activities, %	17.4	19.5	17.8	20.5	17.4	19.5	17.8	20.5
Employed in public sector , %	42.9	38.3	29.2	47.2	34.2	26.3	21.1	31.9

Notes. ^a Dissimilarity index *DI* (known also as *Duncan index*, see Duncan and Duncan 1955) is a number between 0 and 1, with 0 indicating equal distribution of ethnic (or other) groups among sectors, and 1 (or 100%) indicating complete segregation. In the given context, *DI* shows the minimal proportion of non-Latvians which would have to change sector of economic activity in order to make their distribution among sectors identical to that of Latvians'.

Source: Calculation based on LFS data

Table 1.13: Sector of economic activity in the main job by Latvian language skills, 2005.
Employees aged 18-64

	Latvian language skills				Total
	Native	Good	Medium	Poor	
Agriculture, forestry and fishing (A+B)	4.9	1.7	3.3	5.5	4.1
Mining (C)	0.3	0.2	0.2	1.4	0.3
Manufacturing (D)	14.2	15.2	22.8	30.7	16.4
Energy, gas and water supply (E)	2.3	1.0	2.3	0.5	2.0
Construction (F)	6.4	6.8	9.4	11.9	7.2
Trade (G)	17.3	22.4	18.9	9.6	18.1
Hotels and restaurants (H)	3.3	5.8	4.9	4.1	4.0
Transport, storage and communication (I)	6.8	6.8	11.3	13.3	7.7
Financial intermediation (J)	1.6	1.6	0.2	0.0	1.4
Real estate and business activities (K)	7.2	8.8	4.9	9.6	7.4
Public administration (L)	9.5	5.4	2.5	1.8	7.4
Education (M)	13.2	8.4	6.8	5.5	11.0
Health (N)	6.6	7.5	5.1	4.1	6.4
Other services (O+P+Q)	6.3	8.4	7.4	1.8	6.6
Total	100.0	100.0	100.0	100.0	100.0
Dissimilarity index^a (%) between given category and native Latvian speakers					
by fourteen sectors of economic activities					
Agriculture, forestry and fishing (A+B)	4.9	1.7	3.3	5.5	4.1
Industry (C+D+E+F)	23.2	23.2	34.7	44.5	25.9
Market services (G+H+I+J+K)	36.3	45.4	40.2	36.7	38.6
Non-market services (L+M+N+O+P+Q)	35.6	29.7	21.8	13.3	31.4
Total	100.0	100.0	100.0	100.0	100.0
Dissimilarity index^a (%) between given category and native Latvian speakers:					
by four main sectors of economic activities					
Number of observations	2515	811	487	218	4031
<i>Notes.</i> ^a Dissimilarity index <i>DI</i> (known also as <i>Duncan index</i>, see Duncan and Duncan 1955) is a number between 0 and 1, with 0 indicating equal distribution of ethnic (or other) groups among sectors, and 1 (or 100%) indicating complete segregation. In the given context, <i>DI</i> shows the minimal proportion of workers with good, medium and poor Latvian language skills which would have to change sector of economic activity in order to make their distribution among sectors identical to that of native Latvian speakers’. <i>Source:</i> Calculation based on a representative survey of employees implemented in November 2005-January 2006 by <i>RS Group</i> in the framework of the study of determinants of wages commissioned by the Ministry of Welfare for the National Program of Labor Market Studies.					

Table 2.1: Full-Time^a Employees by Educational Attainment, 1997-2005

Education	1997	2002	2005
Higher	19.8	23.0	23.9
Secondary general	23.2	24.6	25.8
Upper secondary vocational ^b	46.3	39.9	38.5
Basic general	9.5	9.9	10.0
Basic vocational	n.a.	1.8	1.3
Less than basic	1.2	0.8	0.5
Total	100.0	100.0	100.0

Note: ^a As defined by respondents, disregarding number of hours worked. When estimating earnings function the sample, however, is restricted to those working at least 35 hours per week.

^b Includes vocational after basic, secondary professional, secondary after vocational, and post-secondary (non-tertiary) vocational education. For 1997 includes also basic vocational (c. a. 3 percent)

Source: Calculation based on LFS data

Table 2.2: Estimated *ceteris paribus* wage differentials^a (%) associated with educational attainment. Full-time^b employees, 2005 and 2002

Education (vs. basic general or less)		All	Men	Women	Latvians	Non- Latvians	Private	Public	Urban	Rural
Higher:										
	2005	76.9	67.8	90.3	85.7	61.0	60.5	109.1	66.9	126.1
	2002	81.5	70.2	99.3	88.5	73.6	69.5	101.1	81.4	91.3
Secondary general										
	2005	14.6	12.5	21.8	17.1	10.4	12.3	20.3	12.3	17.7
	2002	17.3	16.1	23.7	19.6	15.2	17.1	20.4	19.3	11.4
Upper secondary vocational ^c										
	2005	18.5	16.2	24.9	18.4	17.5	14.6	30.4	14.9	28.4
	2002	18.5	16.7	25.8	16.5	22.4	16.4	24.8	20.2	14.0
Basic vocational										
	2005	17.2	11.4	32.9		24.2	10.6 ^d	45.1 ^e	16.6	
	2002									
Higher vs. secondary general										
	2005	54.4	49.2	56.3	58.6	45.8	42.9	73.7	48.6	92.1
	2002	54.7	46.7	61.1	57.7	50.7	44.8	67.0	52.0	71.7

Note: ^a Controls include education, age and its square, gender, ethnicity, marital status, type of contract, tenure and its square, ownership sector, sector of economic activity (14 dummies), job location (5 regions, capital city, rural/urban). All differentials are based on coefficients significantly different from zero (at 1% level in most cases, at 5% level in other cases). Basic vocational education was not significant in 2002. ^b As defined by respondents but excluding those working less than 35 hours per week. ^c Includes vocational after basic, secondary professional, secondary after vocational, and post-secondary (non-tertiary) vocational education. ^d Significant at 12% level. ^e Based on small number of respondents.

Source: Calculation based on LFS data

Table 2.3: Estimated *ceteris paribus* wage differentials^a associated with job tenure. Full-time^b employees, 2005

	All	Men	Women	Latvians	Non-Latvians	Private	Public	Urban	Rural
Mean tenure, years	7.5	6.5	8.5	7.3	7.7	5.9	10.6	7.6	7.0
	Without occupation and plant size controls								
Return to the 1 st year of tenure, %	6.9	6.8	7.6	8.6	4.4	7.3	7.4	8.1	2.0
Return to an extra year of tenure at mean tenure, %	1.2	1.2	1.2	1.1	1.2	1.1	1.2	1.2	1.4
	With occupation and plant size controls								
Return to the 1 st year of tenure, %	5.8	5.1	7.5	6.5	4.9	6.2	7.9	7.1	1.1
Return to an extra year of tenure at mean tenure, %	0.7	0.7	0.6	0.7	0.7	0.6	0.8	0.8	0.8

Note: ^a Controls include education, age and its square, gender, ethnicity, marital status, type of contract, tenure and its square, ownership sector, sector of economic activity (14 dummies), job location (5 regions, capital city, rural/urban).
^b As defined by respondents but excluding those working less than 35 hours per week.
Source: Calculation based on LFS data

Table 2.4: Incidence of fixed-term contracts and estimated *ceteris paribus* wage differentials^a between temporary and permanent workers. Full-time^b employees, 2005 and 2002. Percent

		All	Men	Women	Latvians	Non-Latvians	Private	Public	Urban	Rural
Share of fixed-term workers	2005	6.9	8.7	4.9	7.2	6.4	9.1	2.5	5.7	12.0
	2002	11.5	14.3	8.7	11.0	12.3	14.1	7.5	10.7	15.5
		Without occupation and plant size controls								
Wage differential	2005	-8.8	-8.2	-8.2	-7.9	-11.1	-8.7	-19.3	-9.8	-9.1
	2002	-12.1	-11.6	-11.5	-14.1	-7.8	-13.0	-9.6	-11.6	-11.6
		With occupation and plant size controls								
Wage differential	2005	n/s	n/s	n/s	n/s	n/s	n/s	-14.3	n/s	n/s
	2002	-7.1	-8.0	-4.9	-9.6	n/s	-8.5	-5.7	-7.4	n/s

Note: ^a Controls include education, age and its square, gender, ethnicity, marital status, type of contract, tenure and its square, ownership sector, sector of economic activity (14 dummies), job location (5 regions, capital city, rural/urban). Only differentials which are based on coefficients significantly different from zero are shown (not significant ones are denoted as “n/s”).
^b As defined by respondents but excluding those working less than 35 hours per week.
Source: Calculation based on LFS data

Table 2.5: Gross male – female wage differentials and productivity differentials.

Full-time ^a employees, 2005 and 2002.			Percent	
	Gross wage differential ^b	Productivity differential (without occupation controls) ^c	Productivity differential (with occupation controls) ^d	
			9 major groups of occupations	27 two-digit groups of occupations
2005	24.6	4.6	7.7	11.4
2002	31.2	6.8	10.6	15.6

Note: ^aAs defined by respondents but excluding those working less than 35 hours per week.

^b These differentials, according to conventional methodology of decomposition, are calculated as $\exp(d) - 1$, where d is the difference between mean log net monthly wages of males and females.

^c Differential in (geometric) mean predicted wages of males and females, using earnings function estimated over pooled sample without gender dummy; known also as *explained gender pay gap*. Controls include education, age and its square, ethnicity, marital status, type of contract, tenure and its square, ownership sector, sector of economic activity (14 categories), job location (5 regions, capital city, rural/urban).

^d Additional controls include also plant size (four categories).

Source: Calculation based on LFS data

Table 2.6: Gross wage differentials and productivity differentials between ethnic Latvian and non-Latvian workers.

Full-time ^a employees, 2005 and 2002.			Percent
	Gross wage differential ^b	Productivity differential (without occupation controls) ^c	Productivity differential (with occupation controls) ^d
2005	9.6	0.8	2.6
2002	10.2	-2.1	1.2

Note: ^a As defined by respondents but excluding those working less than 35 hours per week.

^b These differentials, according to conventional methodology of decomposition, are calculated as $\exp(d) - 1$, where d is the difference between mean log net monthly wages of Latvian and non-Latvian workers.

^c Differential in (geometric) mean predicted wages of Latvians and non-Latvians, using earnings function estimated over pooled sample without ethnicity dummy; known also as *explained gender pay gap*. Controls include education, age and its square, gender, marital status, type of contract, tenure and its square, ownership sector, sector of economic activity (14 categories), job location (5 regions, capital city, rural/urban).

^d Additional controls: nine major groups of occupations and plant size (four categories).

Source: Calculation based on LFS data

Table 2.7: Gross wage differentials and productivity differentials between native Latvian speakers and other workers, by self-reported Latvian language skills level.

Full-time ^a employees aged 18-64, 2005.					Percent
Knowledge of Latvian language	Share of workers	Dissimilarity index of occupational segregation ^b from native speakers	Gross wage differential ^c	Productivity differential (without occupation controls) ^d	Productivity differential (with occupation controls) ^e
Native	62.3	-	-	-	-
Good	19.9	7.0	-2.1	-6.0	-5.7
Medium	12.2	24.7	10.0	3.7	3.0
Poor	5.4	49.0	13.4	11.3	12.4

Note: ^a. Full-time workers aged 18 to 64.

^b With respect to nine major groups of occupations. ^c These differentials, according to conventional methodology of decomposition, are calculated as $\exp(d) - 1$, where d is the difference between mean log net monthly wages of native Latvian speakers and other workers.

^d Differential in (geometric) mean predicted wages of native Latvian speakers and the given group, using earnings function estimated over pooled sample without language and ethnic dummies; known also as *explained gender pay gap*. Controls include education, age, gender, marital status, type of contract, tenure and its square, ownership sector, sector of economic activity (14 categories), job location (5 regions, capital city, rural/urban). ^e Additional controls: nine major groups of occupations and firm size (four categories).

Source: Calculation based on *ad hoc* survey implemented by RS Group in the framework of National program of labor market studies (4040 respondents).

Table 2.8: Estimated *ceteris paribus* regional effects on earnings.

Full-time employees, 2005 and 2002.						Percent
Job location	Share of [fulltime] workers			Wage deviations from non-weighted average (based on normalized regression coefficients)		
	2002	2005	Change	2002	2005	Change (percentage points)
Riga	45.0	43.2	-1.8	25.8	21.5	-4.2
Pieriga	11.8	14.4	2.6	8.3	3.7	-4.6
Vidzeme	8.7	8.0	-0.6	-7.9	-7.6	0.2
Zemgale	10.2	10.1	-0.2	-3.1	8.6	11.8
Kurzeme	12.3	12.0	-0.3	1.1	-5.4	-6.4
Latgale	12.1	12.4	0.3	-18.6	-16.5	2.1
Total	100.0	100.0	0.0			

Notes: Results are derived from earnings functions which control for education, age and its square, gender, marital status, type of contract, tenure and its square, ownership sector, sector of economic activity (14 categories), job location (5 regions, capital city, rural/urban). Results are almost identical when occupation and plant size controls are also included.

Table 2.9: Estimated *ceteris paribus* regional wage differentials^a (%)
Full-time^b employees. 2005

Job location ^c	All	Men	Women	Latvians	Non-Latvians	Private	Public	Urban	Rural
Riga	31.6	33.4	26.2	34.8	22.1	33.1	26.1	25.7	-
Pieriga	12.3	17.3	(5.0)	9.6	11.6	11.8	(7.6)	(1.2)	23.3
Zemgale	17.6	24.5	10.9	16.2	19.3	19.8	14.9	14.5	21.2
Kurzeme	(2.5)	(3.6)	(0.4)	(1.4)	(1.8)	(-0.3)	(8.1)	(-2.4)	9.0
Latgale	-9.6	-14.0	(-5.1)	-9.7	-13.3	-15.6	(-1.8)	-14.5	(-2.3)
Rural	-2.5	-5.5	(-0.6)	-5.7	7.1	(2.1)	-10.6	-	-

Note: ^a Controls include education, age and its square, gender, ethnicity, marital status, type of contract, tenure and its square, ownership sector, sector of economic activity (14 categories), job location (5 regions, capital city, rural/urban). Results are almost identical when occupation and plant size controls are also included. Differentials shown in parentheses are based on coefficients which are not significantly different from zero.

^b As defined by respondents but excluding those working less than 35 hours per week.

^c Regions compared to Vidzeme; Rural compared to cities other than Riga.

Source: Calculation based on LFS data

Table 2.10: Estimated *ceteris paribus* regional wage differentials^a (%)
when local unemployment rate is controlled.
Full-time^b employees. 2005

Job location ^c	All	Men	Women	Latvians	Non-Latvians	Private	Public	Urban	Rural
Riga	22.4	24.0	19.8	25.5	11.8	17.2	28.8	14.4	
Pieriga	8.2	13.6	(2.2)	(6.1)	(6.3)	(5.1)	9.1	(-2.7)	20.3
Zemgale	17.8	23.9	11.4	16.2	18.0	19.1	15.2	14.4	20.7
Kurzeme	(2.6)	(3.8)	(0.7)	(1.1)	(1.9)	(-0.1)	8.3 ^d	(-2.0)	(7.4)
Latgale	(-3.9)	-8.1	(-0.6)	(-3.0)	(-8.4)	-7.0 ^d	(-3.1)	-9.3	(6.0)
Rural	(-1.7)	(-3.2)	(-0.3)	-4.4	10.4	4.2 ^d	-10.8		
Wage curve elasticity ^e									
Estimate	-0.110	-0.123	-0.083	-0.112	-0.133	-0.195	(0.028)	-0.140	(-0.098)
Robust s. e. ^f	0.053	0.068	0.053	0.067	0.056	0.062	0.047	0.055	0.135

Note: ^a Controls include education, age and its square, gender, ethnicity, marital status, type of contract, tenure and its square, ownership sector, sector of economic activity (14 categories), job location (5 regions, capital city, rural/urban), and registered unemployment rate by 33 NUTS-4 regions (districts and main cities). Results are almost identical when occupation and plant size controls are also included. Employees working abroad excluded. Differentials shown in parentheses are based on coefficients which are not significantly different from zero. ^b As defined by respondents but excluding those working less than 35 hours per week. ^c Regions compared to Vidzeme; Rural compared to cities other than Riga.

^d Based on coefficients significant at 12% level.

^e Wage elasticity with respect to registered local unemployment rate (in the end of the previous year).

^f Based on White (1984) correction with clustering on NUTS-4 regions.

Source: Calculation based on LFS data.

Table 2.11: Estimated *ceteris paribus* industry effects on earnings.

Full-time^a employees. Latvia, 2005 and 2002.

Percent

Economic activity	Share of [fulltime] workers			Wage deviations from non-weighted average (based on normalized regression coefficients)		
	2002	2005	Change	2002	2005	Change (% points)
Agriculture	3.1	3.1	0.0	-19.3	-23.7	-4.5
Forestry and fishing	4.1	3.7	-0.4	-3.4	2.6	6.1
Manufacturing and mining	20.7	17.8	-2.9	-0.5	-4.1	-3.6
Energy, gas and water supply	3.0	2.8	-0.2	9.4	7.6	-1.8
Construction	7.0	9.4	2.4	9.9	11.3	1.4
Trade	16.0	15.9	0.0	-8.1	-8.7	-0.6
Hotels and restaurants	2.5	2.7	0.3	-7.7	-11.8	-4.1
Transport and communication	9.5	9.5	0.0	5.4	9.4	4.0
Financial intermediation	1.5	1.9	0.3	45.6	38.9	-6.8
Real estate and business activities	3.6	4.6	1.0	1.7	1.5	-0.2
Public administration	8.2	9.2	1.0	11.0	11.6	0.6
Education	8.8	8.6	-0.2	-10.3	-6.8	3.6
Health	6.5	5.9	-0.6	-12.7	-2.8	10.0
Other services	5.5	4.8	-0.8	-6.8	-11.5	-4.7
	100.0	100.0	0.0			

Notes: Results are derived from earnings functions which control for education, age and its square, gender, marital status, type of contract, tenure and its square, ownership sector, sector of economic activity (14 categories), job location (5 regions, capital city, rural/urban).

Source: Calculation based on LFS data.

**Table 2.12: Changes in the earnings of full-time workers
between 2002 and 2005: decomposition analysis (log points)**

	Characteristics effect	Share (%)	Coefficients Effect	Share (%)	Total effect	Share (%)
Age	-0.006	-2.2	-0.077	-28.2	-0.083	-30.5
Education	-0.002	-0.7	-0.026	-9.5	-0.028	-10.2
Gender	0.001	0.5	0.000	-0.2	0.001	0.4
Ethnicity	0.001	0.5	-0.004	-1.3	-0.002	-0.7
Contract type	0.005	1.7	-0.012	-4.5	-0.008	-2.8
Marital status	-0.001	-0.2	-0.006	-2.2	-0.006	-2.3
Tenure	0.005	2.0	0.002	0.8	0.008	2.8
Hours worked	-0.007	-2.7	0.061	22.3	0.053	19.6
Ownership sector	0.003	1.2	0.003	1.1	0.006	2.3
Sector of economic activity	0.007	2.5	0.002	0.8	0.009	3.3
Job location	-0.003	-1.0	-0.040	-14.7	-0.043	-15.7
Months fixed effects	-0.001	-0.5	0.000	-0.1	-0.002	-0.6
Total excl. intercept	0.003	1.1	-0.097	-35.6	-0.094	-34.6
Intercept ^a	0.000	0.0	0.367	134.6	0.367	134.6
Aggregate effect	0.003	1.1	0.269	98.9	0.272	100.0

Notes: Results are derived from earnings functions which control for education, age and its square, gender, marital status, type of contract, tenure and its square, number of hours usually worked per week and its square, ownership sector, sector of economic activity (14 categories), job location (5 regions, capital city, rural/urban).

^a The effect of intercept represents impact of the factors not included in the model.

Source: Calculation based on LFS data.